MEDICINE
WORKBOOK

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MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

Contents

CLINICAL CARDIOLOGY

CHAPTER 1. HEART SOUNDS

PARTS OF STETHOSCOPE

✦ Bell → used for .... pitch sounds.

✦ Low pitched sounds :
- ....
- ....
- ....

✦ Diaphragm → used for .... pitch sounds.

✦ High pitched sounds :
- ....
- ....
- ....
- ....

S1 :

✦ Due to closure of ....

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Abnormalities of S1 :

Loud S1	Soft S1
Hyperdynamic conditions	Low cardiac output
Short PR interval	Prolonged PR

S2 :

- Due to **closure of** ....

Splitting of S2 :

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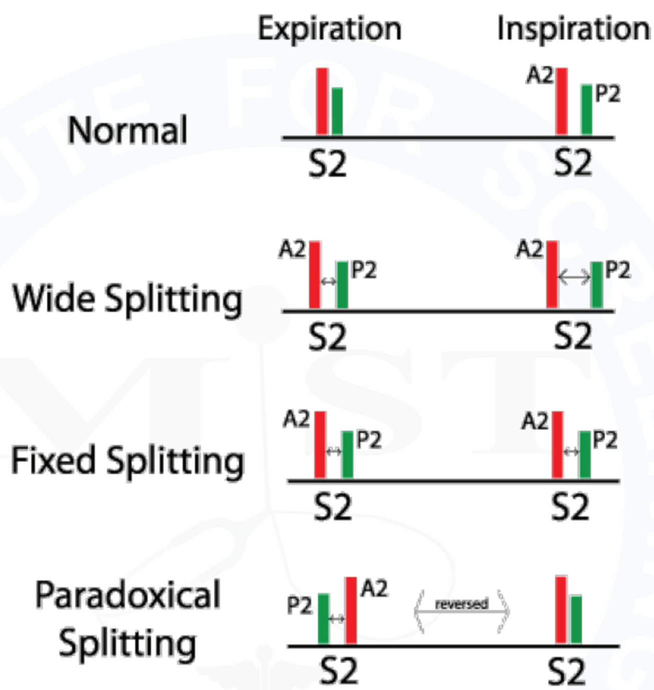
Expiration
Inspiration
Normal
A2
P2
S2
S2
Wide Splitting
A2
P2
S2
A2
P2
S2
Fixed Splitting
A2
P2
S2
A2
P2
S2
Paradoxical Splitting
P2
A2
reversed
S2
S2

S3 :

Physiological S3	Pathological S3
Children	Dilated cardiomyopathy
Pregnancy	Systolic heart failure (HFrEF)
Athletes	

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S4 :

Active space

Workbook Questions

1. The first heart sound (S1) is produced primarily due to:
   A. Closure of aortic valve
   B. Closure of pulmonary valve
   C. Closure of mitral and tricuspid valves
   D. Rapid ventricular filling

2. Which of the following conditions is associated with a soft S1?
   A. Mitral stenosis
   B. Short PR interval
   C. Left bundle branch block
   D. Increased sympathetic tone

3. Physiological splitting of S2 occurs due to:
   A. Early closure of aortic valve during inspiration
   B. Delayed closure of pulmonary valve during inspiration
   C. Fixed delay in pulmonary valve closure
   D. Increased LV stroke volume

4. Wide and fixed splitting of S2 is classically seen in:
   A. Pulmonary hypertension
   B. Left bundle branch block
   C. Atrial septal defect
   D. Aortic stenosis

5. Which condition produces a paradoxical (reversed) splitting of S2?
   A. Right bundle branch block
   B. Atrial septal defect
   C. Left bundle branch block
   D. Pulmonary embolism

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6. S3 gallop is best described as:
A. Sound of atrial contraction
B. Sound during rapid ventricular filling
C. Sound due to AV valve closure
D. Sound due to semilunar valve closure

7. Presence of an S3 in an adult >40 years most commonly indicates:
A. Hypertrophic cardiomyopathy
B. Restrictive cardiomyopathy
C. Volume overload and systolic dysfunction
D. Severe aortic stenosis

8. S4 heart sound is caused by:
A. Rapid ventricular filling
B. Sudden deceleration of blood in dilated ventricle
C. Atrial contraction against a non-compliant ventricle
D. Early closure of AV valves

9. S4 is absent in which of the following conditions?
A. Long-standing hypertension
B. Ischemic heart disease
C. Atrial fibrillation
D. Hypertrophic cardiomyopathy

10. A loud P2 component of S2 suggests:
A. Aortic stenosis
B. Mitral regurgitation
C. Pulmonary hypertension
D. Left ventricular hypertrophy

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CHAPTER 2: PULSE

WHAT IS PULSE?

- Pulse pressure (PP) :

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✦ Pulse deficit (PD) :

Abnormal Pulses :

Pulse	Condition
Catacrotic	Normal
Pulsus parvus	Decreased cardiac output like • Cardiogenic • Neurogenic
Pulsus et tardus	LBBB
Pulsus parvus et tardus	AS
Anacrotic pulse	AS
Water hammer pulse	AR (any condition with wide pulse pressure)
Pulsus bisferiens	HOCM, AR, AR+AS
Dicrotic pulse	Decrease cardiac output
Pulsus alternans	LVF

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Active space

Workbook Questions

1. A slow-rising, low-amplitude pulse (pulsus parvus et tardus) is classically seen in:
   A. Aortic regurgitation
   B. Hypertrophic obstructive cardiomyopathy
   C. Aortic stenosis
   D. Constrictive pericarditis

2. A collapsing (water-hammer) pulse is best explained by:
   A. Increased peripheral resistance
   B. Rapid systolic upstroke with rapid diastolic runoff
   C. Decreased stroke volume
   D. Delayed ventricular ejection

3. Bisferiens pulse is most characteristically associated with:
   A. Pure aortic stenosis
   B. Aortic regurgitation
   C. Severe aortic stenosis with LV failure
   D. Combined aortic stenosis and aortic regurgitation

4. Pulsus alternans most commonly indicates:
   A. Cardiac tamponade
   B. Severe left ventricular systolic dysfunction
   C. Hypertrophic cardiomyopathy
   D. Mitral stenosis

5. Pulsus paradoxus is defined as:
   A. Rise in systolic BP during inspiration
   B. Fall in systolic BP >10 mmHg during inspiration
   C. Fall in diastolic BP during expiration
   D. Equal pulse volume in inspiration and expiration

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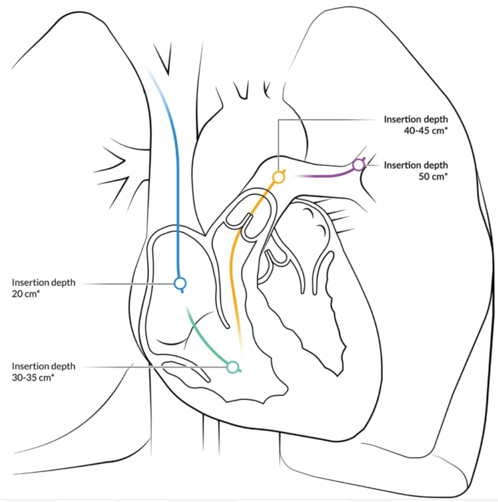
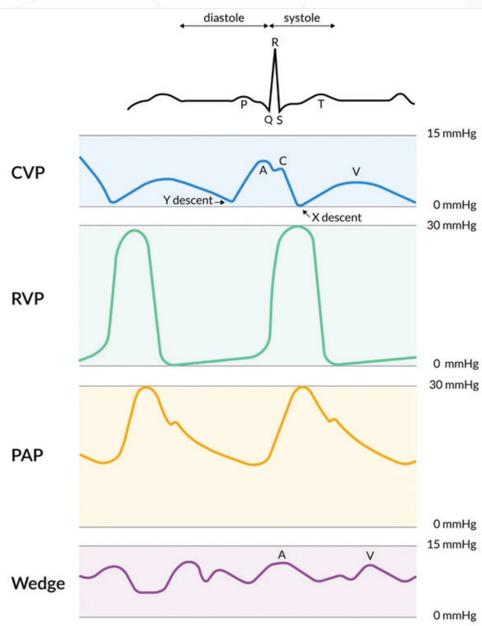
CHAPTER 3: JUGULAR VENOUS PULSATIONS





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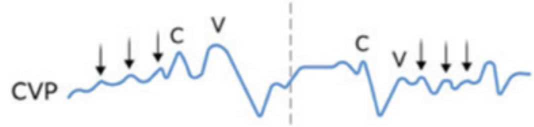
H. Atrial fibrillation/flutter

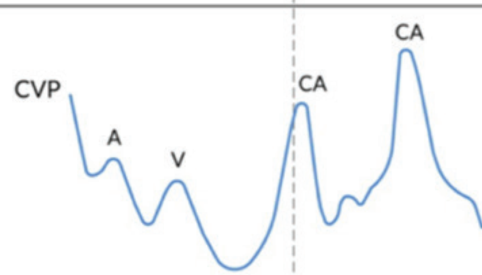
Characteristics:
- *Absence of A-wave;*
- *C and V-waves most prominent peaks;*
- *Beat to beat changing morphologies;*
- *Small amplitude pressure waves (fibrillation/flutter waves)*



I. AV dissociation

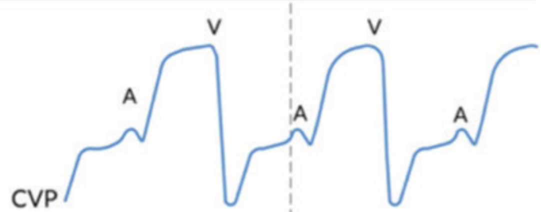
Ventricular tachycardia, AVNRT, Complete heart block

Characteristics:
- *Cannon A-wave during ventricular systole*



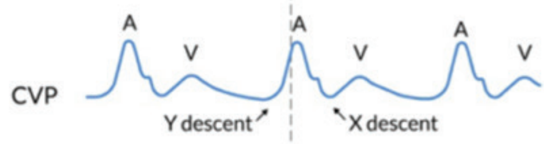
J. Tricuspid insufficiency

Characteristics:
- *Early holosystolig large V-wave;*
- *Merge with the C-wave;*
- *Disappearance of X-descent*



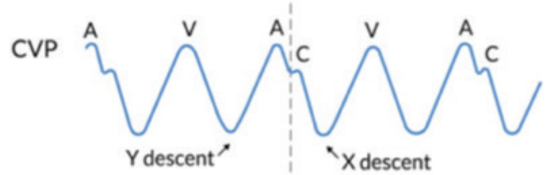
L. Tricuspid valve stenosis

Characteristics:
- *Large A-wave;*
- *Slow Y-descent*



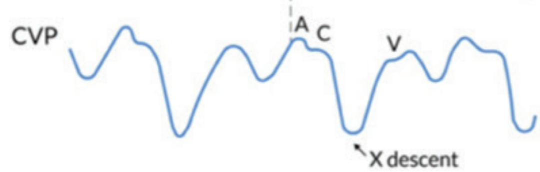
N. Pericardial constriction, RV infarction or restrictive cardiomyopathy

Characteristics:
- *Tall A-wave* - *Steep X-descent*
- *Tall V-wave* - *Steep Y-descent*



O. Tamponade

Characteristics:
- *Tall A-wave*
- *Tall V-wave*
- *Absent or diminished Y-descent*



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Workbook Questions

1. Absent 'a' waves in JVP are classically seen in:
   A. Tricuspid stenosis
   B. Complete heart block
   C. Atrial fibrillation
   D. Pulmonary hypertension

2. Cannon 'a' waves occur due to
   A. Increased RV compliance
   B. Atrial contraction against closed tricuspid valve
   C. Rapid ventricular filling
   D. Severe tricuspid regurgitation

3. Which JVP waveform is prominent in tricuspid regurgitation?
   A. 'a' wave
   B. 'x' descent
   C. 'v' wave
   D. 'y' descent

4. Prominent x descent with absent y descent is most characteristic of:
   A. Constrictive pericarditis
   B. Cardiac tamponade
   C. Right ventricular failure
   D. Tricuspid stenosis

5. Kussmaul's sign is best described as:
   A. Fall in JVP during inspiration
   B. Rise in JVP during expiration
   C. Rise or failure of JVP to fall during inspiration
   D. Prominent 'a' waves during inspiration

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6. A 30-year-old woman presents with fever and acute breathlessness. She has hypotension, raised JVP, muffled heart sounds, and absence of y descent with a prominent x descent. Pulsus paradoxus is present. Most likely diagnosis?

A. Constrictive pericarditis

B. Cardiac tamponade

C. Acute pulmonary embolism

D. Tricuspid regurgitation

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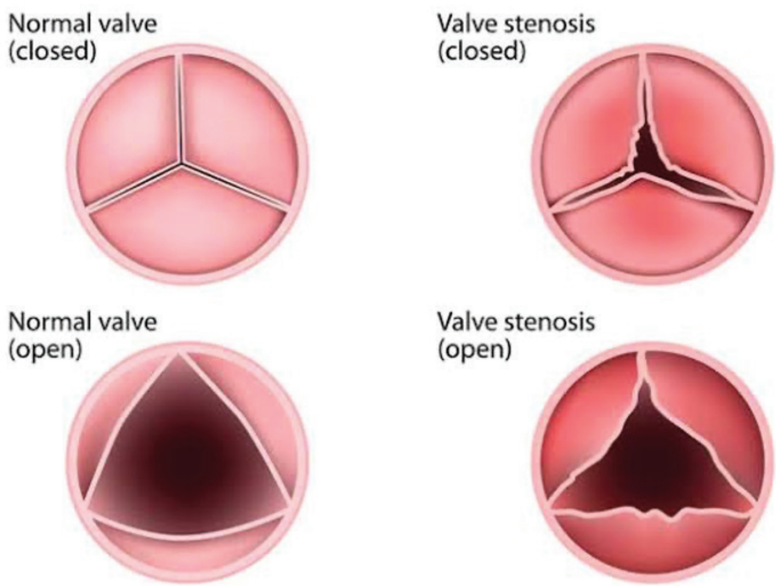
CHAPTER 4: VALVULAR HEART DISEASES

AORTIC VALVE DISEASE:

AV Anatomy



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Normal AV orifice area:

With respect to cardiac cycle, when aortic valve opens?

Active space

Active space

AORTIC STENOSIS:

Causes for AS:

- M/C cause (overall)-
- M/C cause in Young-

Pathophysiology of AS:

Clinical features of aortic stenosis :

- Patients with aortic stenosis can be symptomatic or pseudo symptomatic.
- Classical symptoms :

  1. Angina (earliest)
  2. Syncope
  3. Dyspnea (late symptom indicating severe aortic stenosis)

- Heyde syndrome :

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Median survival in aortic stenosis in patients who develop :

- Angina -
- Syncope -
- Dyspnoea -

Examination findings in AS

- Pulse
- BP
- Apex beat
- Auscultation:

Investigations:

- IOC-
- Severe AS:
- Critical AS:

Management of AS:

- Beta blockers?
- Diuretics?
- Definitive Management?

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✦ Indications for AV replacement:

1. All symptomatic patients.
2. Asymptomatic patients with ejection fraction < 50%.
3. Severe aortic stenosis : undergoing other cardiac surgery, regardless of symptom status / ejection fraction.
4. Abnormal exercise testing (like : Treadmill testing).

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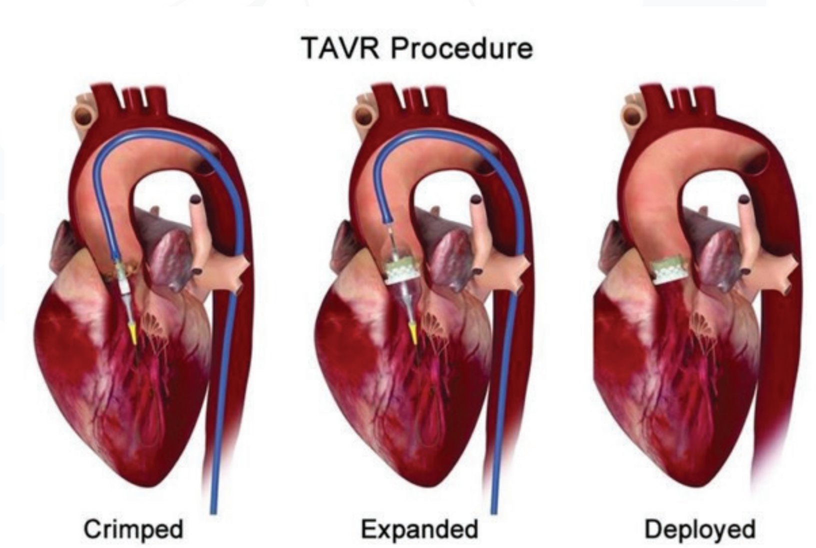
✦ What if the patient is elderly / unfit for open heart surgery?



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AORTIC REGURGITATION:

Causes for AR :

Valvular causes :

- Congenital BAV
- **Endocarditis**
- Rheumatic fever
- Myxomatous degeneration
- Trauma

Root causes :

- HTN
- Marfan syndrome
- Ehler-Danlos syndrome
- Aoritis
- Aortic dissection

Pathophysiology of acute AR:

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Pathophysiology of Chronic AR:

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Clinical features of AR :

A. Pulse -

B. Blood Pressure -

C. Apex beat -

- ✦ .... in acute AR.
- ✦ .... in chronic AR

D. Auscultation

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E. Peripheral signs of AR:

Peripheral signs of AR	
Corrigan sign	Prominent carotid and supraclavicular pulsations
Alfred de Musset sign	Head nodding with heart beat
Landolfi sign	Change in pupil size with heart beat
Muller sign	Uvula pulsation
Becker sign	Retinal artery pulsations
Rosenbach sign	Liver pulsations
Gerhardt sign	Spleen pulsations
Quincke sign	Digital capillary pulsations
Lighthouse sign	Blanching and flushing of the forehead
Traube sign	"Pistol shot"—systolic and diastolic booming sound over femoral artery
Duroziez sign	Intermittent to and fro femoral artery systolic and diastolic murmur generated by light compression with the bell of a stethoscope
Hill sign	Lower limb systolic BP exceeding upper limb systolic BP by more than 20 mm Hg
Water-Hammer pulse	High volume abruptly collapsing pulse
Mayne sign	Diastolic BP drop of >15 mm Hg with arm raised
Lincoln sign	Pulsatile popliteal
Sherman sign	Dorsalispedis pulse unexpectedly prominent in age >75 years

Source : Essentials of Postgraduate Cardiology 2018

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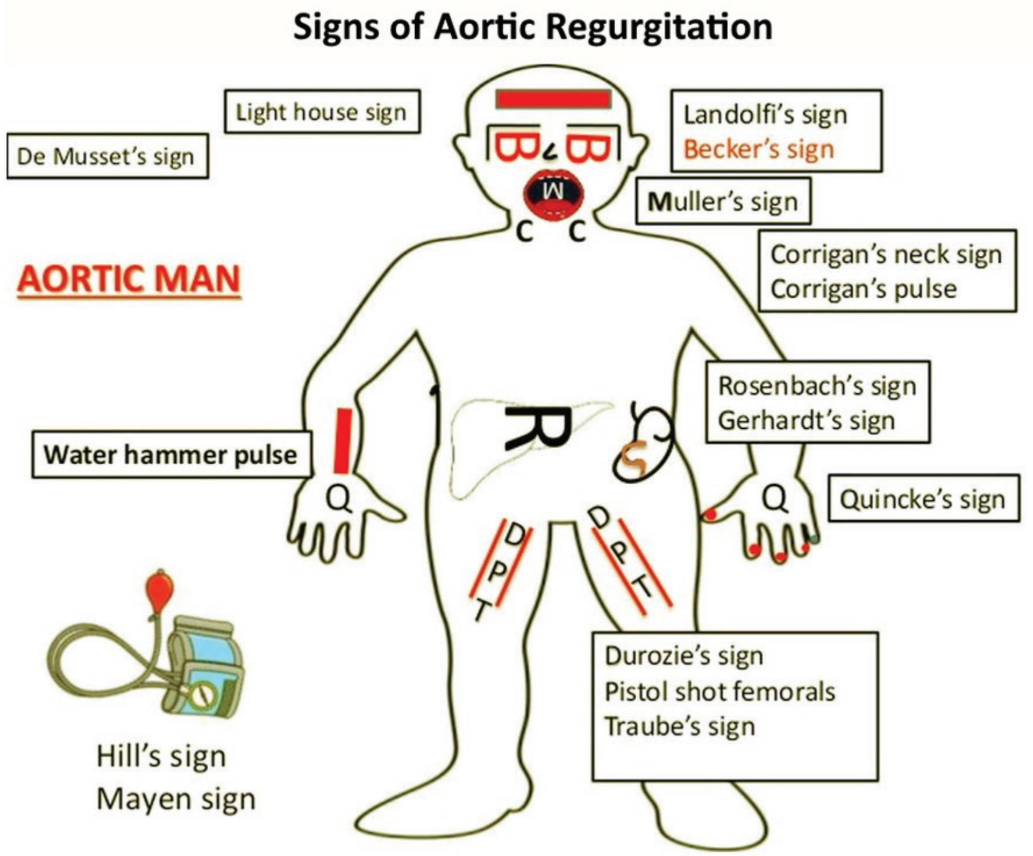

Diagnosis:

- IOC-

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Management:

- Diuretics?
- Nitrates?
- Definitive Management-

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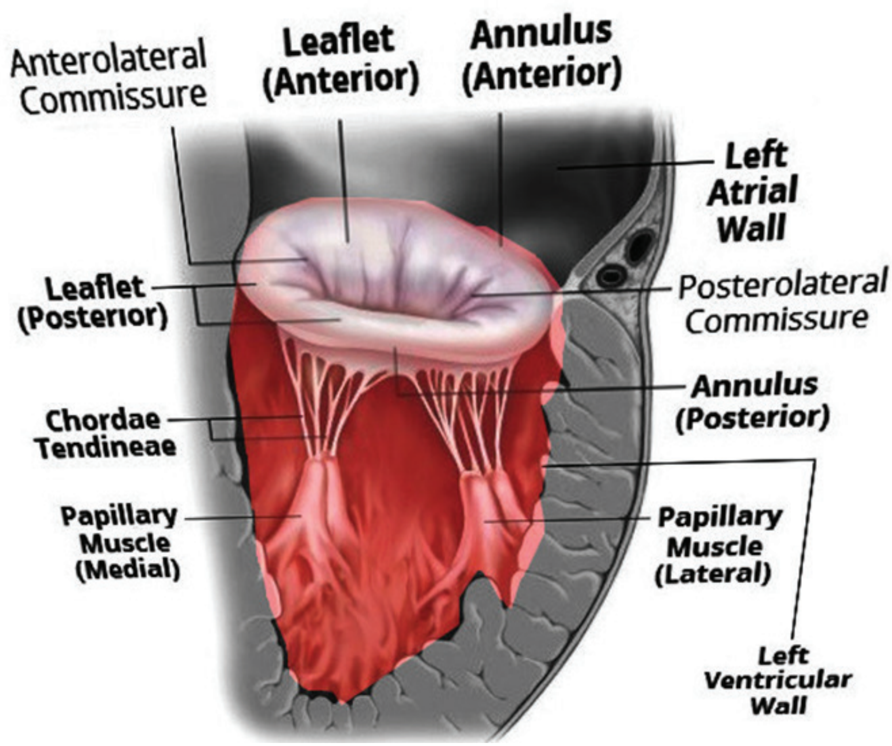
MITRAL VALVE DISEASES

Anatomy of Mitral Valve:



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Normal orifice area:

MITRAL STENOSIS

Causes for mitral stenosis:

- M/C cause

Pathophysiology of MS:

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Clinical Features of MS:

- Earliest symptoms:

Other symptoms-

1. Ortner's syndrome
2. Hemoptysis
3. Cachexia
4. Mitral facies

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Examination findings :

- Pulse -
- BP -
- Apex beat -
- JVP - Increased if ....
- Right parasternal heave & palpable P2 - If ....
- Auscultation

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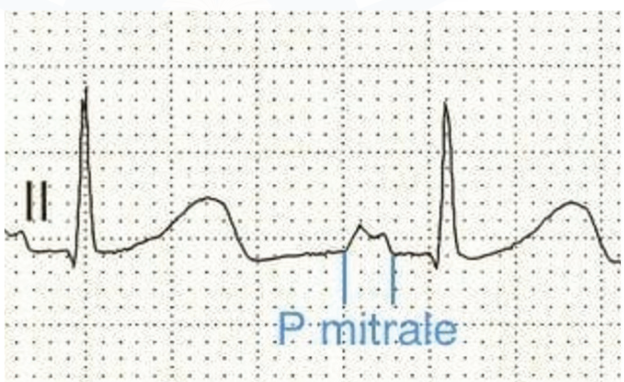
Investigations:

- ECG



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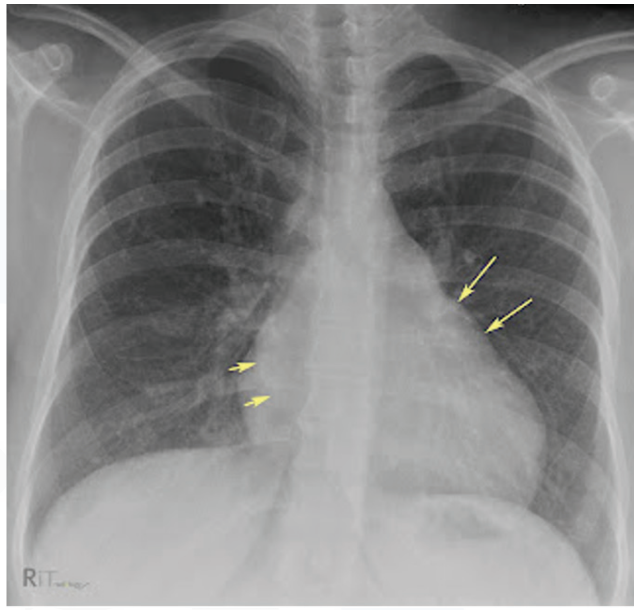
✦ CXR



Echo in MS:

Parameter	Mild	Moderate	Severe
Mean gradient (MG) (mmHg)	<5	5 - 10	> 10
MVA (cm2)	>1.5	1 - 1.5	< 1.0
PASP (mmHg)	<30	30 - 50	> 50
Pressure half time (PHT)			220 ms

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Management of MS:

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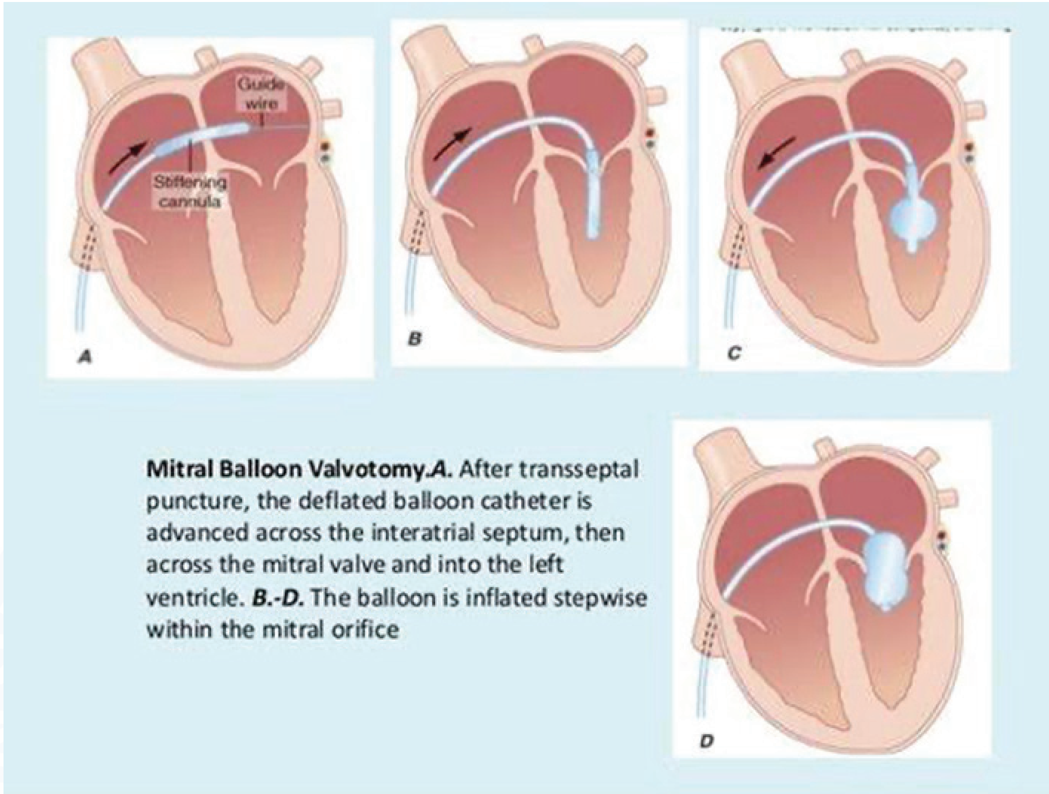

Mitral Balloon Valvotomy. ***A.*** After transseptal puncture, the deflated balloon catheter is advanced across the interatrial septum, then across the mitral valve and into the left ventricle. ***B.-D.*** The balloon is inflated stepwise within the mitral orifice

Indications of BMV - Patient with suitable mitral valve morphology having -

- NYHA class II-IV symptoms with moderate to severe MS.
- Asymptomatic, very severe MS.

  MVA < 1 cm2 + MG > 15 mmHg

Contraindications of BMV

- LLA clot
- > moderate MR
- Densely calcified valve
- Commissural calcification

MITRAL REGURGITATION

Causes of MR :

Valvular disease :

1. RHD
2. **Rupture of chordae** (infective endocarditis)- Cause acute MR
3. Dysfunction of papillary muscle (acute MI)- Cause acute MR
4. Carcinoid

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Secondary MR/ functional MR :

1. DCM
2. IHD
3. Aortic regurgitation

Pathophysiology of acute MR:

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Pathophysiology of Chronic MR:

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Examination findings in chronic MR:

- Pulse
- BP
- Apex beat
- Auscultation:

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	Acute MR	Chronic MR
Presentation	Flash pulmonary edema	Pulmonary congestion
Murmur	Early systolic	Pansystolic
Apex	Non-displaced	Displaced down and out

Investigations:

- ECG
- CXR
- Echo

Management of Acute MR:

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Management of chronic MR:

- Diuretics?
- Beta-blockers?
- Definitive management?

MR is said to be severe on clinical grounds if :

- Hyperdynamic apex beat
- Cardiomegaly
- S_3
- Wide split S2
- Diastolic flow rumble at mitral valve
- Loud P2
- Late systolic (L) parasternal pulsations

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CHAPTER 5: APPROACH TO VALVULAR HEART DISEASE

Active space

Active space

Workbook Questions

1. The most common cause of mitral stenosis in India is:
   A. Degenerative calcification
   B. Congenital
   C. Rheumatic heart disease
   D. Ischemic cardiomyopathy

2. A loud S1 is most characteristic of:
   A. Mitral regurgitation
   B. Mitral stenosis with mobile leaflets
   C. Aortic stenosis
   D. Tricuspid regurgitation

3. Opening snap in mitral stenosis occurs due to:
   A. Sudden closure of mitral valve
   B. Rapid ventricular filling
   C. Sudden tensing of stenotic mitral valve
   D. Turbulent flow across valve

4. A patient with mitral stenosis develops sudden worsening dyspnea. The most likely precipitating event is:
   A. Ventricular tachycardia
   B. Atrial fibrillation
   C. Complete heart block
   D. Pulmonary embolism

5. Which murmur finding suggests severe mitral stenosis?
   A. Long mid-diastolic murmur
   B. Short mid-diastolic murmur
   C. Loud S1
   D. Soft S2

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6. The earliest hemodynamic consequence of mitral regurgitation is:
A. Left ventricular hypertrophy
B. Pulmonary hypertension
C. Left atrial dilation
D. Right ventricular failure

7. A hyperdynamic apex beat is most commonly seen in:
A. Mitral stenosis
B. Aortic stenosis
C. Aortic regurgitation
D. Tricuspid stenosis

8. A collapsing pulse is classically associated with:
A. Mitral regurgitation
B. Aortic stenosis
C. Aortic regurgitation
D. Tricuspid regurgitation

9. Which peripheral sign is NOT associated with aortic regurgitation?
A. Quincke's sign
B. Corrigan pulse
C. De Musset sign
D. Kussmaul sign

10. The most common cause of aortic stenosis in elderly patients is:
A. Rheumatic heart disease
B. Congenital bicuspid valve
C. Degenerative calcification
D. Infective endocarditis

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Clinical Vignettes

11. A 55-year-old man presents with exertional dyspnea and syncope. Examination shows slow-rising pulse, narrow pulse pressure, and ejection systolic murmur radiating to carotids.

Diagnosis?

A. Aortic regurgitation
B. Hypertrophic cardiomyopathy
C. Aortic stenosis
D. Mitral regurgitation

12. A 28-year-old woman with history of rheumatic fever presents with dyspnea. On auscultation: loud S1, opening snap, mid-diastolic murmur at apex.

Most likely diagnosis?

A. Mitral regurgitation
B. Mitral stenosis
C. Tricuspid stenosis
D. Aortic stenosis

13. A patient with acute severe mitral regurgitation is most likely to develop:
A. Chronic pulmonary hypertension
B. Left atrial enlargement
C. Pulmonary edema
D. Right heart failure

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14. Which valvular lesion is most likely to cause atrial fibrillation earliest?
A. Mitral regurgitation
B. Mitral stenosis
C. Aortic stenosis
D. Aortic regurgitation

15. A holosystolic murmur radiating to axilla is typical of:
A. Tricuspid regurgitation
B. Ventricular septal defect
C. Mitral regurgitation
D. Aortic stenosis

16. Which finding favors functional tricuspid regurgitation?
A. Rheumatic fever history
B. Isolated RV dilation
C. Left-sided valvular disease
D. Congenital anomaly

17. A pansystolic murmur that increases with inspiration suggests:
A. Mitral regurgitation
B. Aortic stenosis
C. Tricuspid regurgitation
D. Ventricular septal defect

18. Which valvular lesion is most commonly associated with hemoptysis?
A. Mitral regurgitation
B. Mitral stenosis
C. Aortic stenosis
D. Tricuspid stenosis

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19. A patient with aortic regurgitation has an Austin Flint murmur. This murmur is best described as:
A. Functional mitral stenosis
B. Severe mitral regurgitation
C. Tricuspid stenosis
D. Pulmonary regurgitation

18. Indication for valve intervention in asymptomatic severe aortic stenosis includes:
A. Normal LV function
B. LV ejection fraction <50%
C. Mild symptoms
D. Age <40 years

Answer : 1 - C, 2 - B, 3 - C, 4 - B, 5 - A, 6 - C, 7 - C, 8 - C, 9 - D, 10 - C, 11 - C, 12 - B, 13 - C, 14 - B, 15 - C, 16 - C, 17 - C, 18 - B, 19 - A, 20 - B

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CHAPTER 6: ACUTE RHEUMATIC FEVER

WHAT IS ACUTE RHEUMATIC FEVER?

Clinical features of ARF:

- Carditis
- Arthritis
- Sydenham's chorea

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- Erythema marginatum

- Subcutaneous nodules

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Diagnosis of ARF:

MAJOR CRITERIA	
Low-Risk Populations	**Moderate- and High-Risk Populations**
Carditis (clinical or subclinical[†])	Carditis (clinical or subclinical)
Arthritis (polyarthritis only)	Arthritis (including polyarthritis, monoarthritis, or polyarthralgia[‡])
Chorea	Chorea
Erythema marginatum	Erythema marginatum
Subcutaneous nodules	Subcutaneous nodules
MINOR CRITERIA	
Low-Risk Populations	**Moderate- and High-Risk Populations**
Polyarthralgia	Monoarthralgia
Fever (≥38.5°C)	Fever (≥38°C)
ESR ≥60 mm in the first hour and/ or CRP ≥3.0 mg/dL	ESR ≥30 mm in the first hour and/ or CRP ≥3.0 mg/dL[§]
Prolonged PR interval, after accounting for age variability (unless carditis is a major criterion)	Prolonged PR interval, after accounting for age variability (unless carditis is a major criterion)

Diagnosis of ARF:

- ✦ *Symptomatic*

- ✦ *Eradication of GAS infection*

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✦ Secondary prevention

	AHA/ACC	WHO
No carditis/ valvulitis	5 yrs / 21 yrs of age	5 yrs /18 yrs of age
Carditis only	10 yrs / 21 yrs of age	10 yrs / 25 yrs of age
Valvulitis	10 yrs / 40 yrs of age (Lifelong : if patient had severe carditis/ heart failure)	Life long

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CHAPTER 7: ISCHEMIC HEART DISEASE

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SPECTRUM OF IHD:

Angina:

- Site
- Nature
- Aggravating & relieving factors
- Radiation

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Acute Coronary Syndromes

Approach to ACS:

Role of Biomarkers:

General management of all types of ACS (UA/ NSTEMI/ STEMI):

- Bed rest
- Continuous ECG monitoring
- Antiplatelets- single agent or dual?

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- Pain control:

- Nitrates:

- Beta Blockers:

- Statins:

- Oxygen?

Management of NSTEMI:

- What is definitive management of STEMI?

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- Management approach to STEMI:

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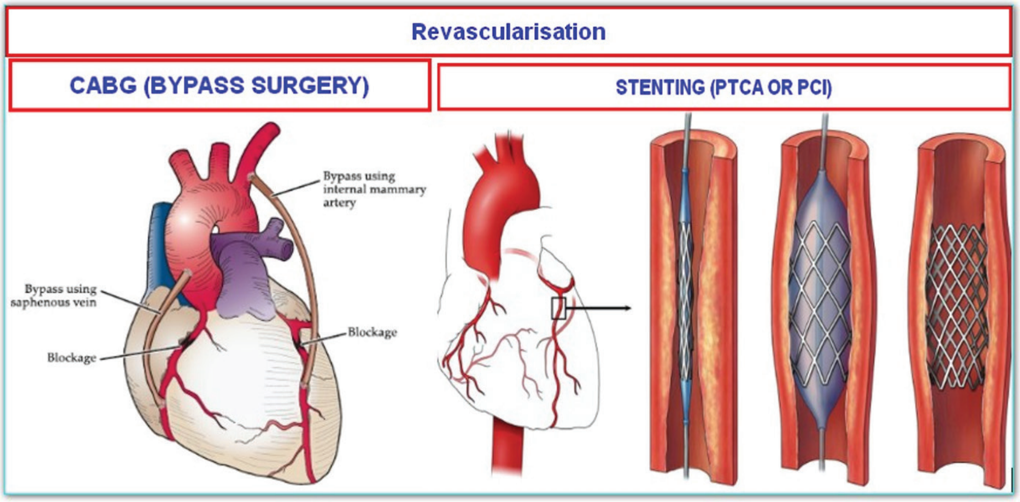

Management of NSTEMI:

Management of Unstable Angina:

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CHAPTER 8: ACUTE PERICARDITIS

Pericarditis on the basis of duration of inflammation is of 3 types :

1. Acute :
2. Subacute:
3. Chronic :

Causes for acute pericarditis:

- Infectious
- Non-infectious

Clinical features of acute pericarditis:

- Chest pain:
- Examination:

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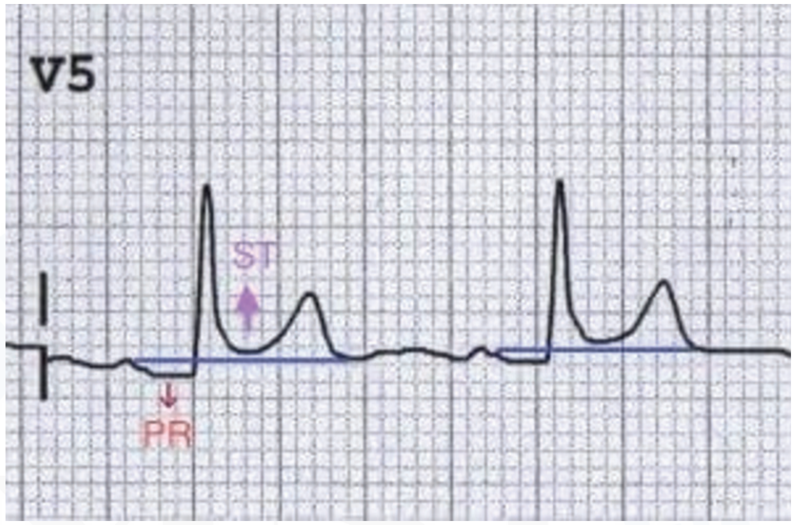
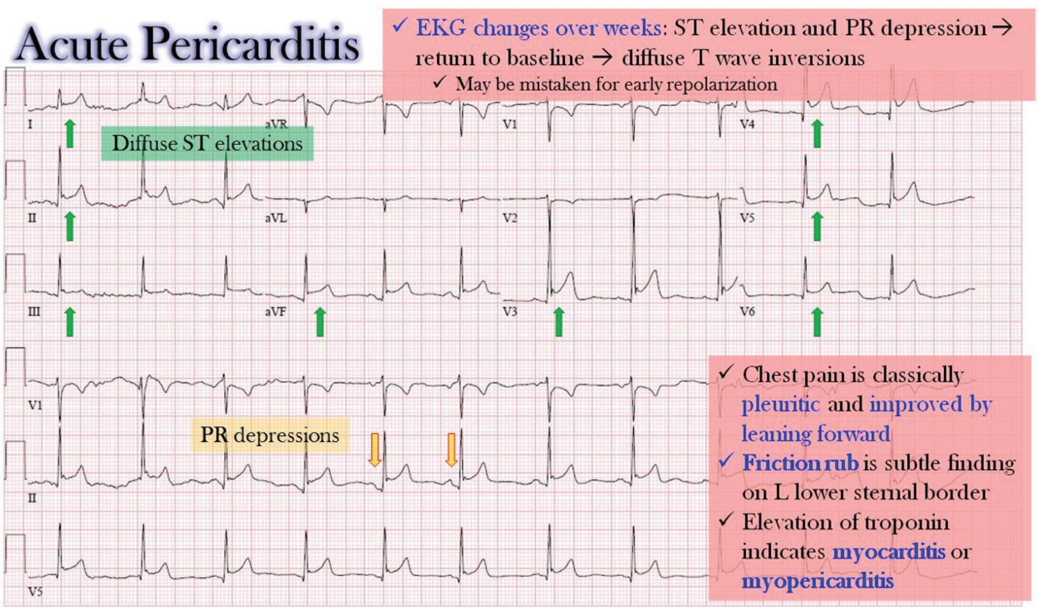
Diagnosis:

✦ IOC-





Active space

Management:

- 1ST line
- 2nd line
- If frequent recurrence –

Active space

Workbook Questions

1. A 60-year-old diabetic presents 2 hours after severe chest pain. ECG shows ST elevation in V1–V4. Most likely artery involved?
   A. RCA
   B. LCX
   C. LAD
   D. PDA

2. The most effective reperfusion strategy for STEMI presenting within 90 minutes is:
   A. Thrombolysis
   B. Primary PCI
   C. Anticoagulation only
   D. Dual antiplatelet therapy

3. A 55-year-old man develops chest pain while climbing stairs. Pain lasts 5 minutes and is relieved by rest. ECG at rest is normal. Most likely diagnosis?
   A. Unstable angina
   B. NSTEMI
   C. Stable angina
   D. Variant angina

4. A 62-year-old man with a history of diabetes presents with acute onset chest pain radiating to the left arm for the last 3 hours. He feels light-headed and nauseated.

   On examination:

   - Pulse: 56/min
   - BP: 86/60 mmHg

Active space

- Jugular venous pressure is elevated
- Lungs are clear
- Heart sounds are normal, no murmur

ST elevation is greater in lead III than lead II.

He becomes more hypotensive after receiving sublingual nitroglycerin

Most likely diagnosis is:

A. Acute anterior wall MI
B. Acute left ventricular failure
C. Inferior wall MI with right ventricular infarction

6. A 28-year-old man presents with acute onset central chest pain for 2 days. The pain is sharp and pleuritic, worsens on lying supine, and is relieved by sitting up and leaning forward. He had an upper respiratory tract infection one week ago.

On examination:

- Temperature: 38°C
- Pulse: 96/min
- BP: 118/76 mmHg
- A scratchy, superficial, triphasic sound is heard along the left sternal border, best heard when the patient leans forward

ECG shows:

- Diffuse concave ST elevation in multiple leads
- PR segment depression

Cardiac troponins are normal.

Most likely diagnosis is:

A. Acute myocardial infarction
B. Unstable angina
C. Acute pericarditis
D. Pulmonary embolism

Active space

MIST

CHAPTER 9: CHRONIC CONSTRICTIVE PERICARDITIS

WHAT IS CCP?

What are the causes for CCP?

- M/C/C in India
- M/C/C worldwide
- Other causes

Active space

Clinical features of CCP:

Symptoms :

- JVP
- Congestive hepatomegaly(Jaundice, Right upper quadrant tenderness)
- Pedal edema
- Exertional dyspnea
- Orthopnea/PND

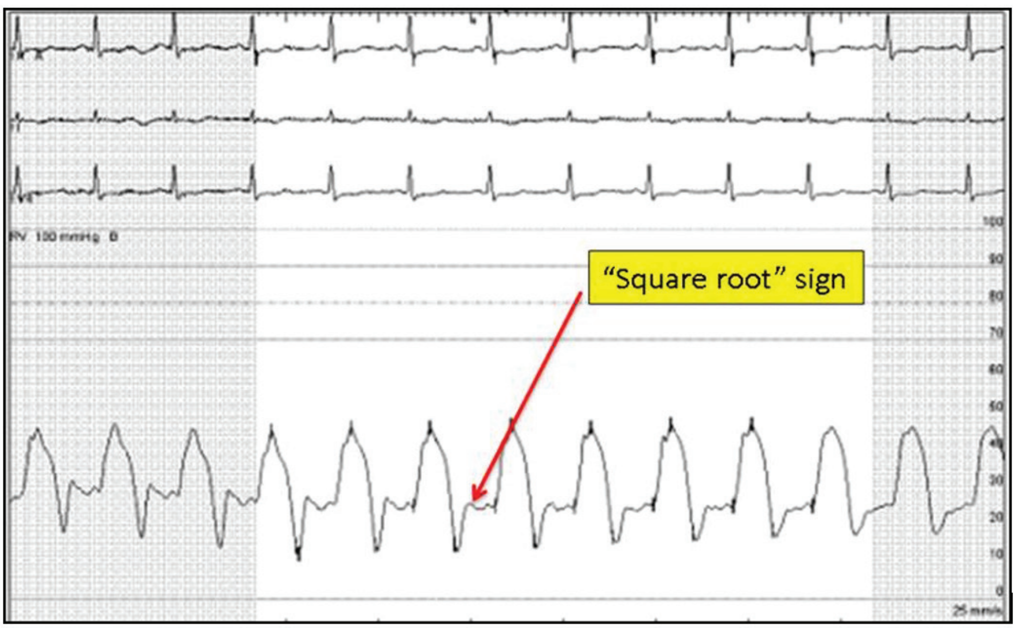
Signs :

- JVP -
- Kussmaul's sign
  - No inspiratory decrease in JVP.
  - It can also be seen in Tricuspid valve stenosis, Restrictive cardiomyopathy and Right ventricular MI.
- Broadbent sign - 11th and 12th rib spaces posteriorly retract during systole.
- Pericardial knock - Diastolic sound due to abrupt cessation of diastolic filling.

Active space

Diagnosis of CCP:

- CXR
- Cardiac catheterization



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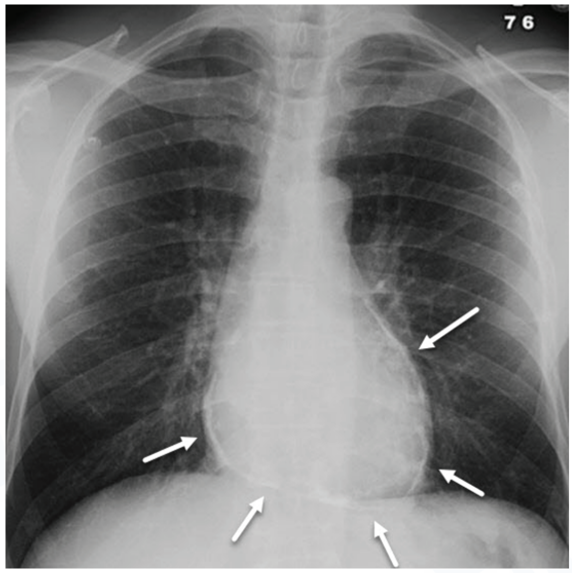


Management of CCP:

- If TB
- If Non-tuberculous

Active space

CHAPTER 10: CARDIAC TAMPONADE

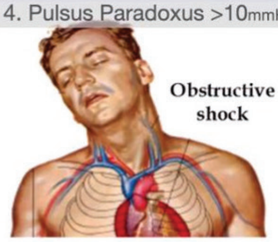
WHAT IS CARDIAC TAMPONADE?

Causes for cardiac tamponade:

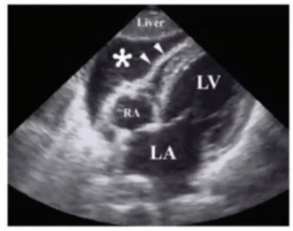
Clinical features:

Active space

Clinical: Beck's Triad	ECG Triad *	POCUS Triad
1. Hypotension	1. Sinus tachycardia	1. Pericardial fluid
2. Jugular vein distension	2. Low voltage	2. RV diastolic collapse
3. Muffled heart sounds	3. Electrical alternans	3. Dilated IVC
4. Pulsus Paradoxus >10mmHg		

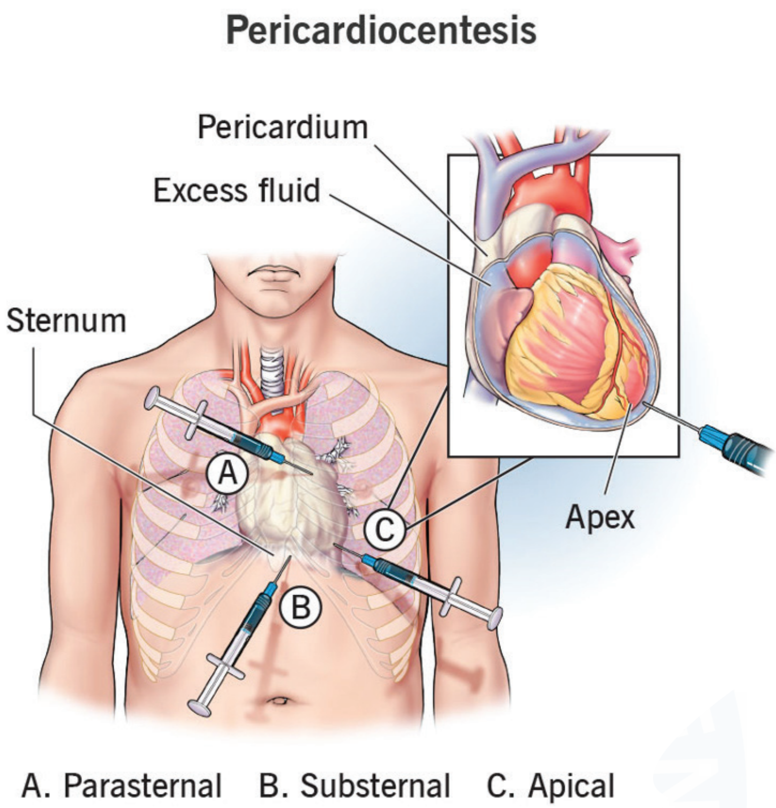


* Indicative of massive pericardial effusion but not always tamponde



Management:

Active space



Active space

Workbook Questions

1. The most important pathophysiological mechanism causing hypotension in cardiac tamponade is:
   A. Reduced systemic vascular resistance
   B. Impaired ventricular filling
   C. Reduced myocardial contractility
   D. Acute valvular regurgitation

2. Which classical triad is associated with acute cardiac tamponade?
   A. Chest pain, fever, tachycardia
   B. Hypotension, raised JVP, muffled heart sounds
   C. Hypotension, bradycardia, clear lungs
   D. Raised JVP, loud S2, pulmonary edema

3. Pulsus paradoxus in cardiac tamponade is best defined as:
   A. Fall in diastolic BP during inspiration
   B. Rise in systolic BP during expiration
   C. Fall in systolic BP >10 mmHg during inspiration
   D. Equal pulse volume in respiration

4. A 35-year-old man presents with fever and acute breathlessness. BP is 80/60 mmHg, JVP is elevated, heart sounds are muffled, lungs are clear. JVP waveform shows prominent x descent with absent y descent.
   Most likely diagnosis?
   A. Constrictive pericarditis
   B. Acute pulmonary embolism
   C. Cardiac tamponade
   D. Restrictive cardiomyopathy

5. The immediate definitive management of cardiac tamponade is:
   A. High-dose NSAIDs B. Diuretics
   C. Pericardiocentesis D. Pericardiectomy

Active space

6. Which of the following drugs should be avoided in acute cardiac tamponade?
A. IV fluids
B. Vasopressors
C. Diuretics
D. Oxygen

7. A 48-year-old man presents with progressive dyspnea and pedal edema over 6 months. Examination shows raised JVP with prominent y descent and rise in JVP during inspiration.
Most likely diagnosis?
A. Cardiac tamponade
B. Restrictive cardiomyopathy
C. Constrictive pericarditis
D. Dilated cardiomyopathy

8. Kussmaul's sign is classically seen in:
A. Cardiac tamponade
B. Acute MI
C. Constrictive pericarditis
D. Severe mitral regurgitation

9. A patient with raised JVP has rapid y descent, pericardial knock, hepatomegaly, and ascites. Echocardiography shows normal ventricular systolic function. Diagnosis?
A. Cardiac tamponade
B. Constrictive pericarditis
C. Restrictive cardiomyopathy
D. Right ventricular infarction

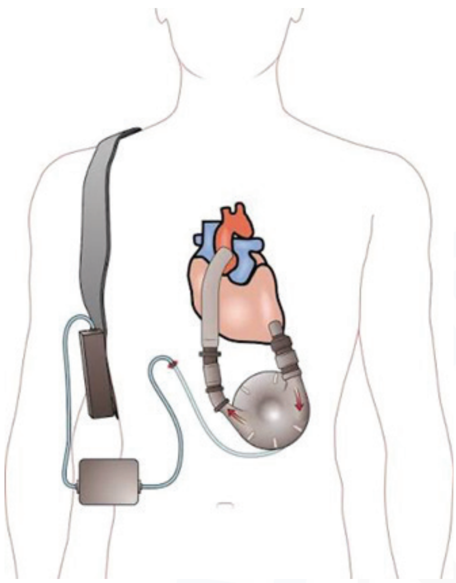
Active space

10. The definitive treatment for chronic constrictive pericarditis is:
A. NSAIDs and colchicine
B. Diuretics alone
C. Pericardiectomy
D. Pericardiocentesis

CHAPTER 11. HEART FAILURE

Active space

Active space



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CHAPTER 12. PULMONARY HYPERTENSION

WHAT IS PULMONARY HYPERTENSION?

What is pulmonary hypertension?

Long term consequences of pulmonary hypertension

Causes for pulmonary hypertension

Active space

Diagnosis:

Management of PH:

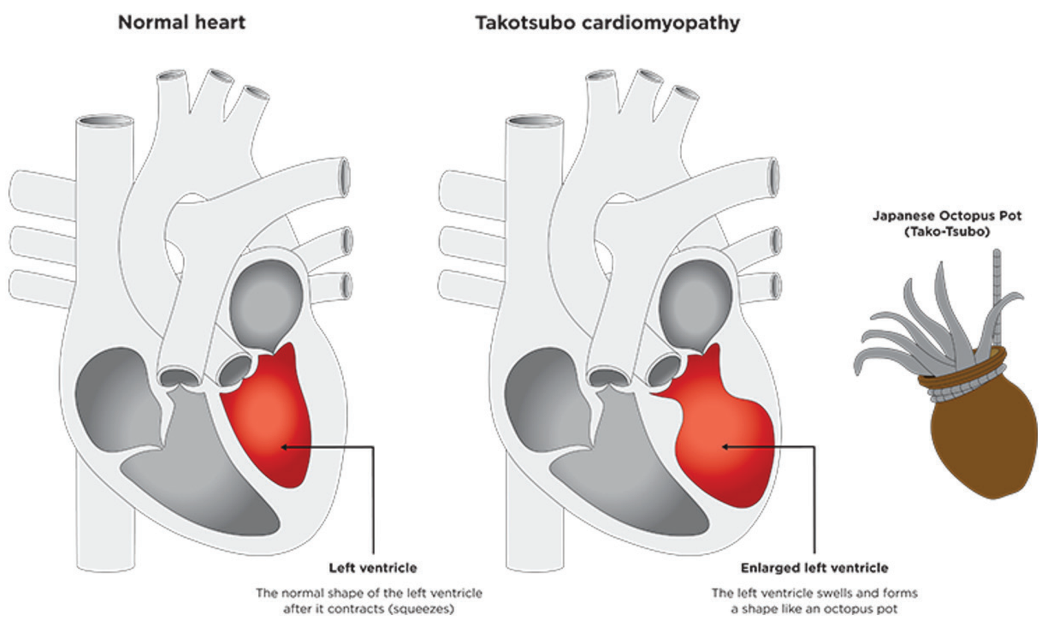
Active space

CHAPTER 13: CARDIOMYOPATHIES

Active space

Active space

Active space


Normal heart
Takotsubo cardiomyopathy
Japanese Octopus Pot
(Tako-Tsubo)
Left ventricle
The normal shape of the left ventricle
after it contracts (squeezes)
Enlarged left ventricle
The left ventricle swells and forms
a shape like an octopus pot

Active space

Workbook Questions

1. The most common type of cardiomyopathy worldwide is:
   A. Hypertrophic cardiomyopathy
   B. Restrictive cardiomyopathy
   C. Dilated cardiomyopathy
   D. Arrhythmogenic RV cardiomyopathy

2. Dilated cardiomyopathy is characterized by:
   A. Diastolic dysfunction only
   B. Small LV cavity
   C. Systolic dysfunction with ventricular dilation
   D. Increased LV wall thickness

3. Which of the following is a reversible cause of dilated cardiomyopathy?
   A. Duchenne muscular dystrophy
   B. Alcoholic cardiomyopathy
   C. Idiopathic cardiomyopathy
   D. Peripartum cardiomyopathy after 1 year

4. A patient with DCM is at highest risk of:
   A. Sudden death due to VT/VF
   B. Mitral stenosis
   C. Pulmonary embolism
   D. LV outflow obstruction

5. The earliest abnormality in hypertrophic cardiomyopathy is:
   A. Reduced ejection fraction
   B. LV cavity dilation
   C. Diastolic dysfunction
   D. Right ventricular failure

Active space

6. Hypertrophic cardiomyopathy is most commonly inherited as:
A. Autosomal recessive
B. Autosomal dominant
C. X-linked
D. Mitochondrial

7. A 22-year-old athlete collapses during exercise. ECG shows LVH. Echo reveals asymmetric septal hypertrophy.

Most common cause of death?
A. Heart failure
B. Stroke
C. Ventricular arrhythmia
D. Mitral regurgitation

8. The murmur of hypertrophic obstructive cardiomyopathy:
A. Decreases with standing
B. Decreases with Valsalva
C. Increases with standing
D. Radiates to carotids

9. Drug of choice for symptomatic HCM is:
A. Digoxin
B. ACE inhibitors
C. Beta blockers
D. Nitrates

10. Which drug is contraindicated in HOCM?
A. Metoprolol
B. Verapamil
C. Digoxin
D. Disopyramide

Active space

11. Restrictive cardiomyopathy is characterized by:
A. Dilated ventricles
B. Impaired systolic function
C. Impaired ventricular filling with normal systolic function
D. LV outflow obstruction

12. The most common cause of restrictive cardiomyopathy worldwide is:
A. Amyloidosis
B. Sarcoidosis
C. Endomyocardial fibrosis
D. Hemochromatosis

13. A patient has ascites, hepatomegaly, raised JVP, preserved EF, and bi-atrial enlargement.
Most likely diagnosis?
A. Dilated cardiomyopathy
B. Constrictive pericarditis
C. Restrictive cardiomyopathy
D. Pulmonary hypertension

14. Which finding favors restrictive cardiomyopathy over constrictive pericarditis?
A. Kussmaul sign
B. Pericardial calcification
C. Biatrial enlargement
D. Prominent y descent

15. Arrhythmogenic right ventricular cardiomyopathy (ARVC) is characterized by:
A. Fibrofatty replacement of RV myocardium
B. RV hypertrophy
C. LV dilation
D. Mitral valve prolapse

Active space

16. ARVC is classically associated with:
A. Inferior MI
B. Ventricular tachycardia with LBBB morphology
C. Atrial fibrillation
D. Complete heart block

17. Takotsubo cardiomyopathy is precipitated by:
A. Chronic alcohol intake
B. Viral infection
C. Emotional or physical stress
D. Pregnancy

18. A key feature distinguishing Takotsubo cardiomyopathy from MI is:
A. ST elevation
B. Elevated troponin
C. Normal coronary arteries
D. Chest pain

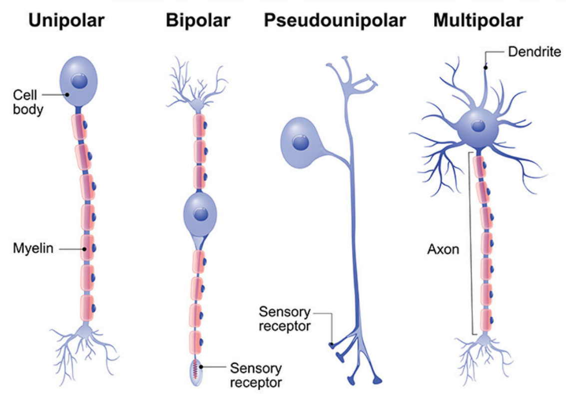
19. Most appropriate management of dilated cardiomyopathy with EF <35% includes:
A. Calcium channel blockers
B. ICD implantation
C. High-dose diuretics alone
D. Digoxin monotherapy

20. Which cardiomyopathy has the worst prognosis?
A. Hypertrophic cardiomyopathy
B. Takotsubo cardiomyopathy
C. Restrictive cardiomyopathy
D. Alcoholic cardiomyopathy

Active space

CLINICAL NEUROLOGY

CHAPTER 1: BASIC FUNCTIONAL NEUROANATOMY

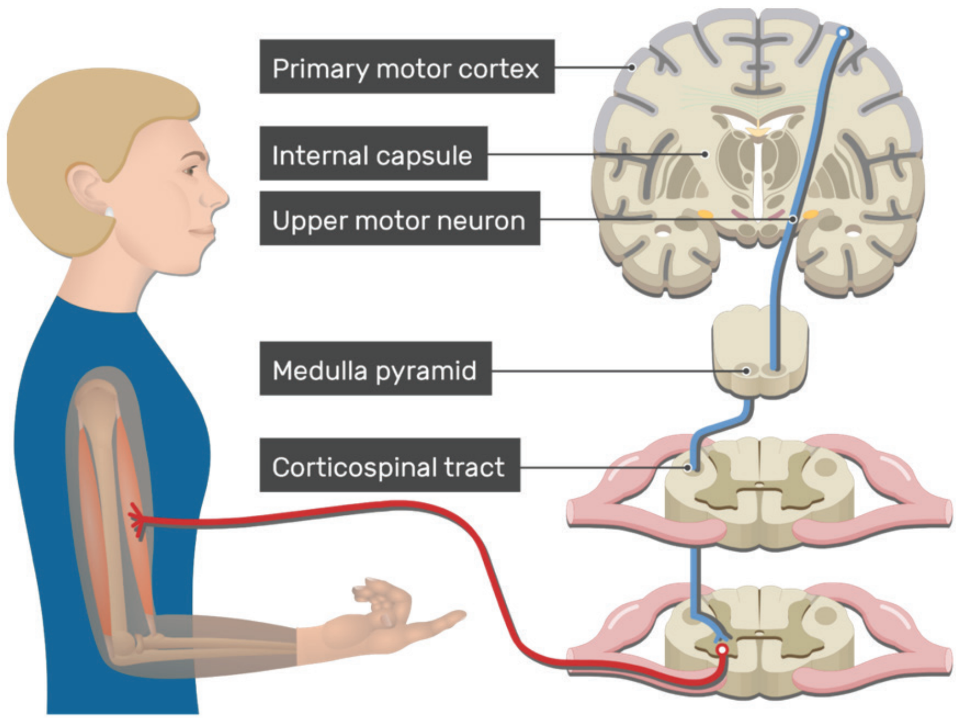


Anatomical organization of nervous system:

Functional organization of nervous system:

Active space

Active space



Functions of LMN:

Functions of UMN:

Active space

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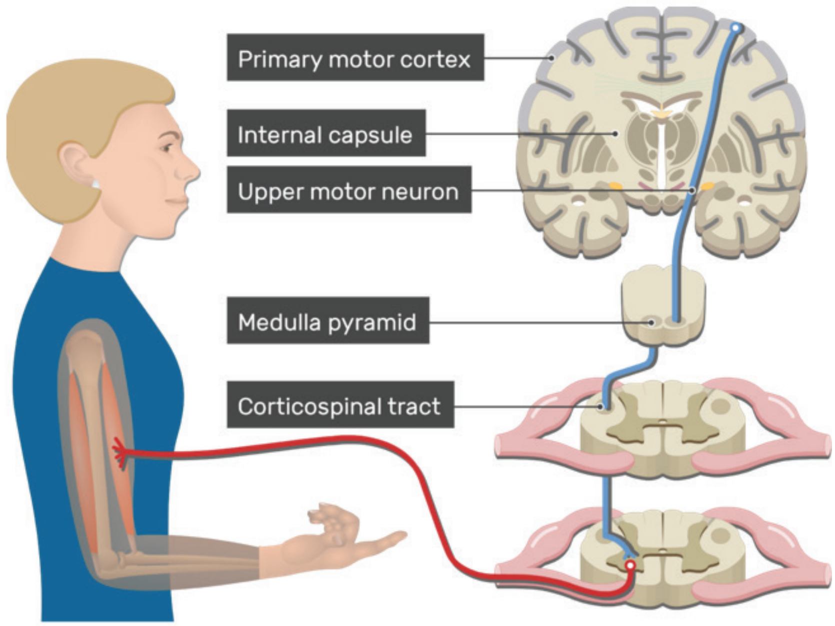


Muscle Disorders:

- Examples

- Weakness-

Active space

- Examination:
    - Strength
    - Tone
    - DTRs
    - Plantar Reflex

NMJ Disorders:

- Examples

- Weakness-

- Examination:
    - Strength
    - Tone
    - DTRs
    - Plantar Reflex

Active space

Peripheral Nerve Disorders:

- Examples

- Weakness-
  - Plus

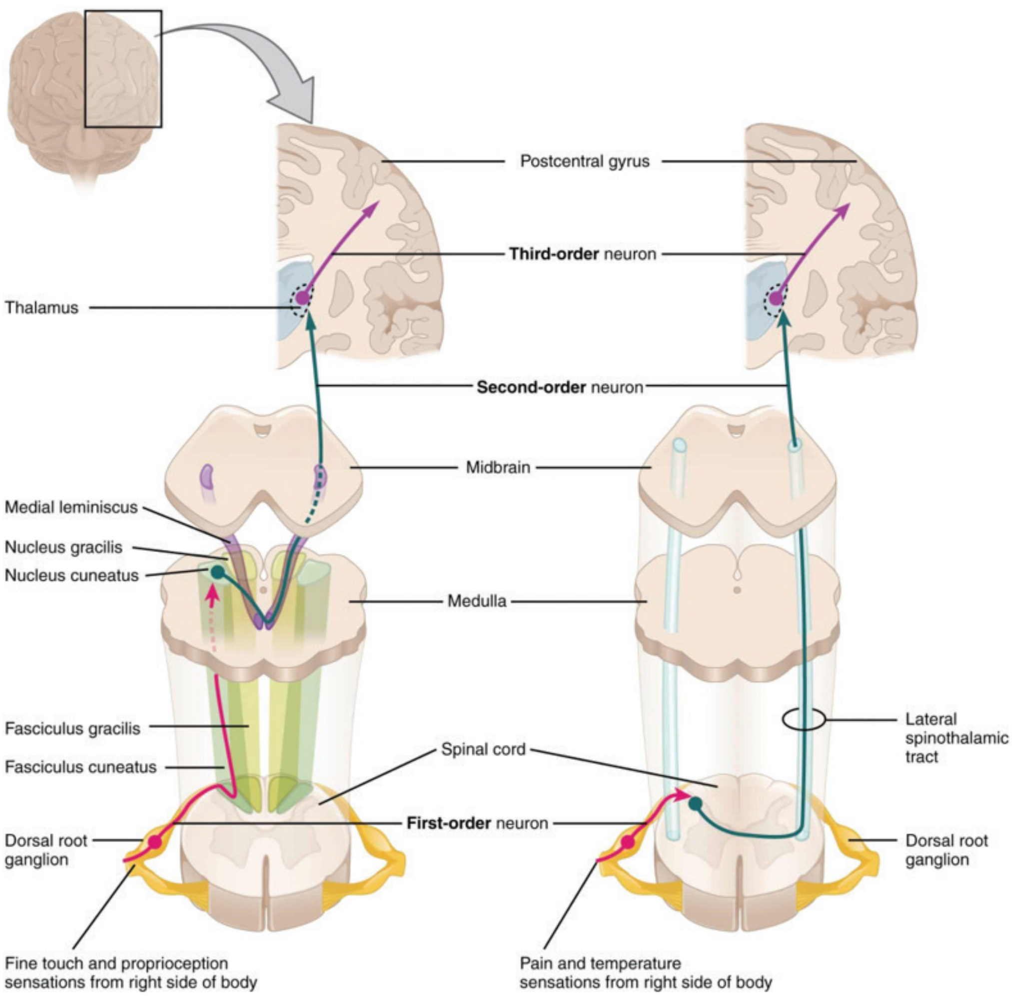
- Examination:
  - Strength
  - Tone
  - DTRs
  - Plantar Reflex

Active space

Active space

Spinal Cord Disorders:

Active space



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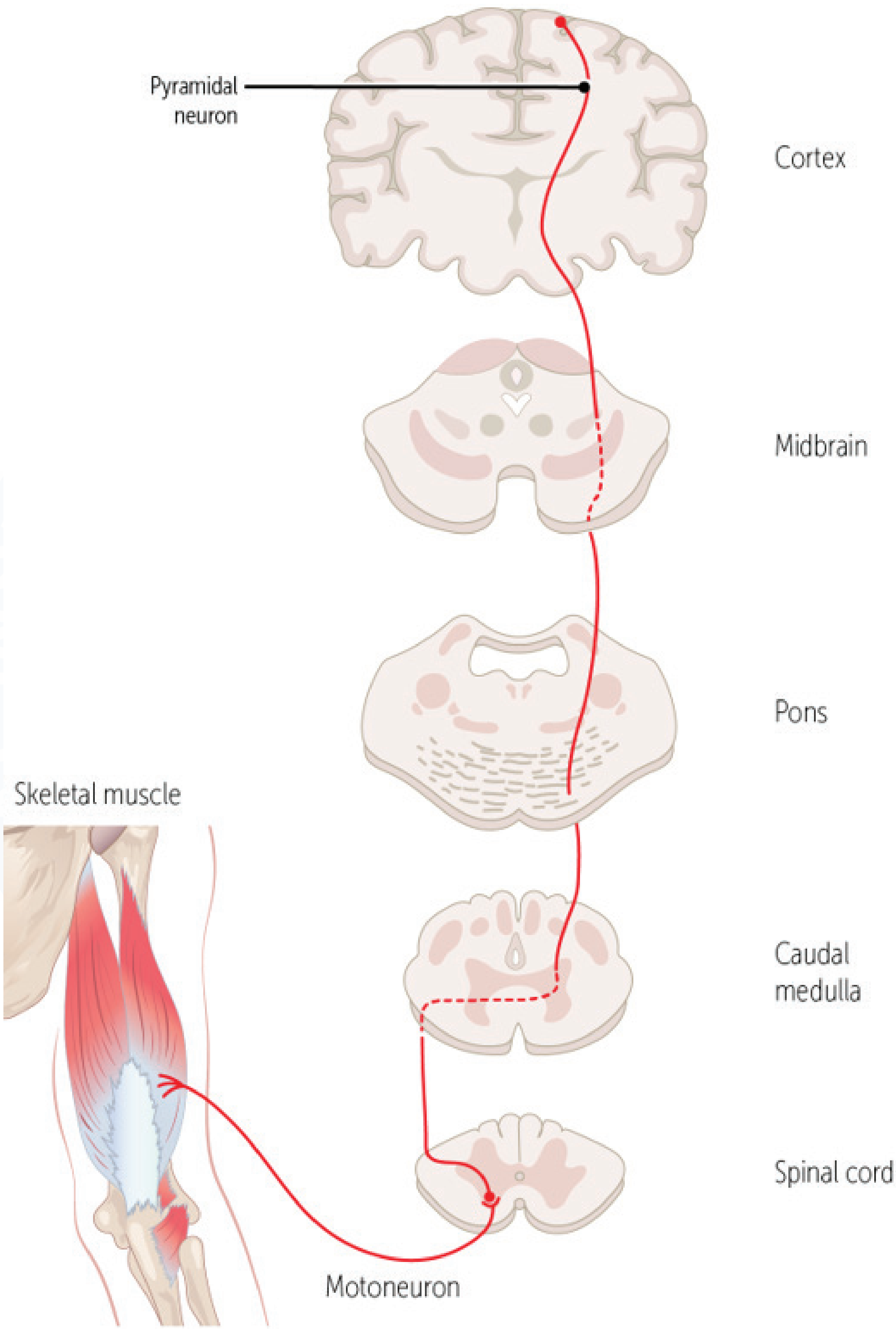

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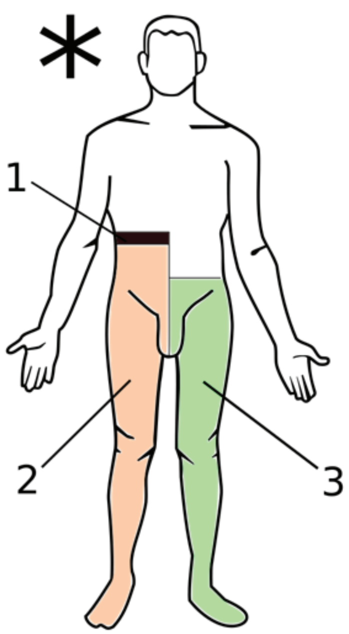
Complete Transection of Spinal Cord:

Hemisection of Spinal Cord:



Active space

Compressive Myelopathies:

Active space

Brain Stem Lesions:

Sub Cortical & Cortical Stroke:

Active space

CHAPTER 3: MYASTHENIA GRAVIS

Review of Basic NMJ Physiology:

Active space

Myasthenia Gravis:

Definition:

Risk factors:

Clinical features:

- Sequence of involvement of muscles:

Active space

- Earliest symptom-

- Other clinical features-
  - o .... muscle weakness
  - o Facial drooping

- Diagnosis:
  - o Most sensitive investigation
  - o Most specific investigation
  - o Other investigations-

Active space

Antibody	Type of Myasthenia Gravis	Remarks
Anti-AChR		
Anti-MUSK		
Anti-LP4R		

✦ Management of MG

Active space

Parameter	Myasthenic Crisis	Cholinergic Crisis
Pathology		
Pupils		
Sludge Symptoms		
Tensilon Test		

Lambert-Eaton Syndrome:

Active space

Workbook Questions

1. A 24-year-old woman presents with ptosis and diplopia that worsen toward evening and improve after rest. Limb strength is normal in the morning.

   Most likely diagnosis?
   A. Lambert-Eaton syndrome
   B. Myasthenia gravis
   C. Botulism
   D. Multiple sclerosis

2. The fundamental pathophysiological abnormality in myasthenia gravis is:
   A. Reduced acetylcholine synthesis
   B. Presynaptic calcium channel blockade
   C. Autoantibodies against postsynaptic acetylcholine receptors
   D. Motor neuron degeneration

3. The most common antibody detected in generalized myasthenia gravis is:
   A. Anti-MuSK
   B. Anti-VGCC
   C. Anti-acetylcholine receptor
   D. Anti-titin

4. The MOST SENSITIVE investigation for diagnosing myasthenia gravis is:
   A. Repetitive nerve stimulation
   B. Edrophonium test
   C. Anti-ACh receptor antibody assay
   D. Single-fiber electromyography (SFEMG)

Active space

5. The BEST confirmatory test for myasthenia gravis is:
A. Ice pack test
B. Repetitive nerve stimulation
C. Anti-acetylcholine receptor antibody assay
D. SFEMG

6. A patient presents with proximal muscle weakness. Reflexes are preserved, and weakness worsens with repeated use.

Most likely diagnosis?

A. Lambert-Eaton syndrome
B. Myasthenia gravis
C. Polymyositis
D. Botulism

7. A 32-year-old man with myasthenia gravis undergoes CT chest showing an anterior mediastinal mass.

Most likely diagnosis?

A. Lymphoma
B. Thymoma
C. Germ cell tumor
D. Retrosternal goiter

8. The drug of choice for symptomatic treatment of myasthenia gravis is:
A. Prednisolone
B. Azathioprine
C. Pyridostigmine
D. IV immunoglobulin

Active space

9. A patient with known MG presents with acute respiratory distress, shallow breathing, inability to lift head, and rising $PaCO_2$.

Best immediate management?

A. Increase pyridostigmine dose
B. High-dose oral steroids
C. Endotracheal intubation + IVIG / plasmapheresis
D. Stop anticholinesterase drugs

10. Which of the following is a clear indication for thymectomy in myasthenia gravis?
A. Pure ocular MG
B. Anti-MuSK-positive MG
C. Thymoma-associated MG
D. Elderly patient with mild generalized MG

Active space

CHAPTER 4: GB SYNDROME

Definition:

Clinical Features:

Classification of GBS Based on NCS:

Active space

GBS Variant	Axonal or Demyelination type	NCS findings	Antibody	Prognosis
AIDP				
AMAN				
AMSAN				
Miller Fischer				

GBS with CNS involvement:- ....

Diagnosis of GBS:

Active space

Management of GBS:

- Most common cause of death in GBS-
- How to differentiate between GBS and extramedullary spinal cord compression?

Active space

Workbook Questions

1. A 34-year-old man presents with acute onset diplopia, unsteady gait, and absent deep tendon reflexes following a diarrheal illness.

   Most likely diagnosis?

   A. Myasthenia gravis
   B. CIDP
   C. Miller-Fisher syndrome
   D. Brainstem stroke

2. The classical triad of Miller-Fisher syndrome includes all EXCEPT:
   A. Ophthalmoplegia
   B. Ataxia
   C. Areflexia
   D. Muscle wasting

3. The antibody most commonly associated with Miller-Fisher syndrome is:
   A. Anti-GM1
   B. Anti-GQ1b
   C. Anti-MAG
   D. Anti-ACh receptor

4. CSF finding in Miller-Fisher syndrome typically shows:
   A. Pleocytosis with normal protein
   B. Normal CSF
   C. Albuminocytologic dissociation
   D. Reduced glucose

Active space

5. The treatment of choice for Miller–Fisher syndrome is:
A. High-dose steroids
B. IV immunoglobulin or plasmapheresis
C. Long-term azathioprine
D. Pyridostigmine

6. A 52-year-old man presents with progressive symmetric proximal and distal weakness evolving over 4 months, with sensory loss and reduced reflexes. Most likely diagnosis?
A. GBS
B. Miller–Fisher syndrome
C. CIDP
D. ALS

7. The key feature distinguishing CIDP from GBS is:
A. Presence of cranial nerve palsy
B. Albuminocytologic dissociation
C. Time course >8 weeks
D. Autonomic dysfunction

8. Nerve conduction studies in CIDP typically show:
A. Axonal degeneration
B. Demyelinating features
C. Neuromuscular junction defect
D. Myopathic pattern

9. Which of the following is an effective long-term treatment for CIDP?
A. IVIG
B. Corticosteroids
C. Plasmapheresis
D. All of the above

Active space

10. Which feature favors CIDP over Miller-Fisher syndrome?
A. Areflexia
B. Ophthalmoplegia
C. Chronic progressive course
D. Antecedent infection

CHAPTER 5: STROKE

DEFINITION:

- An episode of focal/global neurological dysfunction of .... origin lasting for >24 hours or until death.

- Neurodeficit resolves in <24 hours without objective evidence of infarct- ....

Risk stratification for TIA with ABCD2 score

ABCD²	Criteria	Points
Age	≧ 60 years	1
Blood pressure	≧140/80	1
Clinical features	Unilateral weakness	2
	Speech impairment without weakness	1
Duration of Sx	>60minutes	2
	10-59 minutes	1
Diabetes	Yes	1

Score	2day-risk for stroke	Recurrence within 90days
0-3	Low	1.0%
4-5	Moderate	4.1%
6-7	High	8.1%

JAMA 2000;284:2901-2906

Active space

CAUSES FOR STROKE :

1. Ischemic Stroke (85%)
   - Thrombotic Stroke (50%)
   - Thromboembolic Stroke (30-35%) - Patients of Atrial Fibrillation are at higher risk.

2. Hemorrhagic Stroke (15%)
   - Intraparenchymal (IPH) - M/c site of bleed ....
   - Subarachnoid Hemorrhage

ICH Score	Points
GCS score *	
3–4	2
5–12	1
13–15	0
ICH volume **	
≥ 30 cm^3	1
< 30 cm^3	0
IVH ***	
Yes	1
No	0
Infratentorial origin of ICH	
Yes	1
No	0
Age	
≥ 80	1
< 80	0
ICH Total Score	0-6

The ICH Score is a clinical grading scale that allows for risk stratification of patients presenting with ICH. The 5 categories are independent predictors of 30-day mortality. Mortality rises as the ICH Score increases. The use of the ICH Score could improve standardization of treatment protocols and clinical research studies in ICH.

* **GCS** on initial presentation (or after resuscitation)

****ICH volume**, volume on initial CT calculated using ABC/2 method

*** **IVH**, presence of any IVH on initial CT

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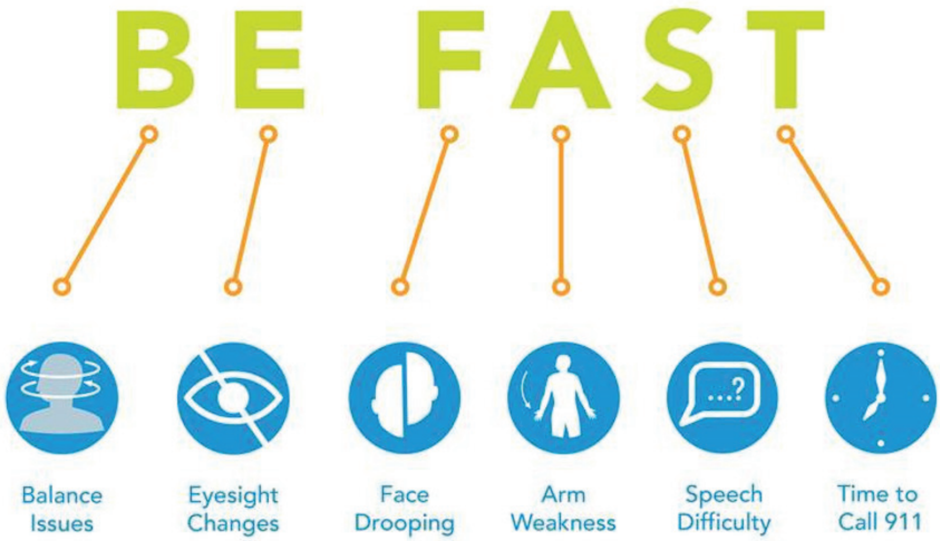
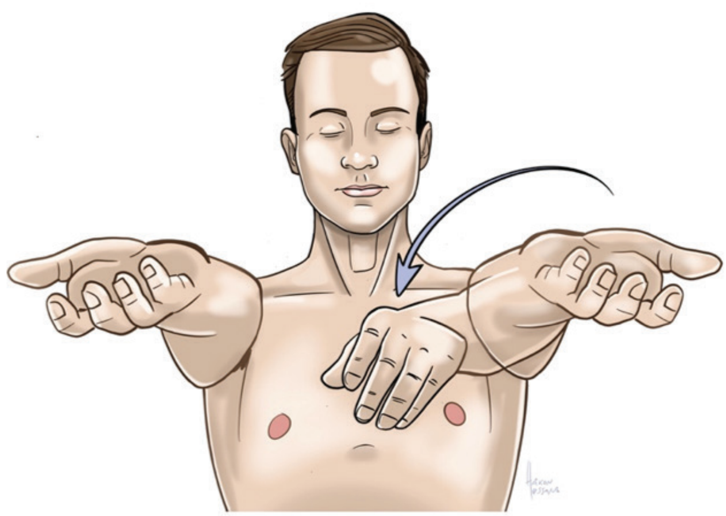
EARLY RECOGNITION OF STROKE :



Pronator drift

1st downward arm drift
2nd forearm pronation
3rd flexion of the wrist and elbow



Approach to a case of suspected stroke:

ISCHEMIC STROKE:

Risk factors for acute ischemic stroke :

- Most common → Hypertension
- Strongest → Diabetes mellitus
- Maximum risk reduction with treatment → AF treated with warfarin

Active space

STROKE PATHOPHYSIOLOGY :

- Cerebral blood flow < 18 mL / 100 gm / min → affected neuronal electrical activities (Penumbra)
- Cerebral blood flow < 8 mL / 100 gm / min → Infarct
- Aim of treatment →..

Goals of stroke treatment :

1. Minimize disability by reducing deficit
2. Reduce mortality
3. Prevent recurrence of stroke

Active space

MANAGEMENT OF ISCHEMIC STROKE:

- In the ER -

1. ABC - Airway, breathing & circulation assessment.

2. GCS (Glasgow coma score)

- GCS <........................→ Intubate to prevent ....

3. BP

- Acute lowering only if SBP > 220 & DBP > 130 mmHg.
- If patient is to be taken for thrombolysis, treat if BP > 185/110.

4. Drugs :

- DAPT (Aspirin + clopidogrel)
- Do not give antiplatelets if the patient is to be taken up for ....

Active space

Definitive Management of acute ischemic stroke:

Active space

CHAPTER 6: HEADACHE

Classification of Headache (ICHD-3)

How to differentiate primary and secondary headaches?

Red flag signs
• Age: >55 years • Fever • Neck stiffness • Focal deficits • Seizures • Projectile vomiting • Bradycardia • Changing character • Worst headache ever

Active space

1. Tension type of headache (TTH) :
   - Gender – F > M
   - Age – **middle** age.
   - Lateralization – **bilateral** (only headache which is bilateral).
   - Localization – **forehead,** parietal & occipital regions.
   - Severity – mild to moderate.

   This localization and severity described by the patient as a ....

   - Duration of episodes:
   - Clues: **vise-like** grip, **band-like** sensation, dull aching, **life stressors.**

Management of TTH

- Acute
- Preventive

2. Migraine :
   - Age –
   - Gender –
   - Lateralization –
   - Localization –

Active space

- Severity –
- Duration of episodes: 4 to 72 hours

Clues :

1. Aura +/-

- **Reversible** abnormal sensory or motor perceptions that patients experience **before the** onset of headache is known as aura.
- Aura lasts for .... minutes (reversible).
- When aura is present it is called .... migraine.

I. Typical aura: Visual aura most common

II. Atypical aura:

- Brainstem dysfunction
- Hemiplegia
- Retinal migraine
- When aura is absent it is called non classical migraine.

2. **Photophobia** & phonophobia
3. Nausea / vomiting
4. Increase headache by **physical activities** such as walking, climbing stairs etc.

Active space

Phases of Migraine :

Migraine progresses through phases :

Diagnosis of migraine :

I. Migraine without aura :

A. At least > 5 attacks fulfilling criteria given below.
B. Headache attacks lasting **4 to 72 hrs** (untreated or unsuccessfully treated).
C. Headache has at least > 2 of the following four characteristics:
  - Unilateral location
  - Pulsating quality
  - Moderate or severe pain intensity
  - Aggravation by or causing avoidance of routine physical activity (eg: walking or climbing stairs).

D. During headache at least >1 of the following:

- Nausea and/or vomiting
- Photophobia and phonophobia.

E. Not better accounted for by another ICHD-3 diagnosis.

II. Migraine with aura:

A. At least ≥ 2 fulfilling criteria given below.

B. ≥1 of the following fully reversible aura symptoms:

- Visual
- Sensory
- Speech and/or language
- Motor
- Brainstem
- Retinal

C. ≥ 3 of the following six characteristics:

- At least one aura symptom spread gradually over ≥ 5 minutes.
- ≥ 2 aura symptoms occur in succession.
- Each individual aura symptom lasts 5 to 60 minutes.
- At least one aura symptom is unilateral.
- At least one aura symptom is positive (eg: feel touching without any touch).
- The aura is accompanied, or followed within 60 minutes, by headache.

D. Not better accounted for by another ICHD-3 diagnosis

Active space

Management of Migraine:

Trigeminal Autonomic Cephalgias:

Characteristics :

1. Unilateral primary headaches
2. Episodes last for < 4 hrs
3. Associated with autonomic symptoms such as
    - Miosis
    - Conjunctival redness
    - Increased lacrimation
    - Nasal stuffiness
    - Rhinorrhea

Active space

Classification of TACs :

	Cluster Headache	Paroxysmal Hemicrania	SUNCT/SUNA
M : F ratio	M > F (3 :1)	F > M	F > M
Duration of attacks			
Frequency of attack per day	Upto 8 attacks/ day	Upto 40 attacks/day	Upto 200 attacks/day
Potential clues	• Triggered by alcohol • Circadian (same time of day) & circannual (same time every year) clustering		• Cutaneous triggers without refractory period
Management	Acute treatment : Preventive treatment : 1. Short term prevention - ----- -------------------- ----------(DOC) 2. Long term prevention - ----- ------------------- ----------- (DOC), Galcanezumab (MAB against CGRP)	Preventive treatment : Excellent response to Indometh-acin	Acute treatment : -------------------- ------- injections Preventive treatment - ------------------- -- (DOC)

Workbook Questions

1. A 24-year-old woman has recurrent unilateral throbbing headaches associated with photophobia, phonophobia, and nausea, lasting 6–12 hours. She prefers to lie in a dark room. Most likely diagnosis?
   A. Tension-type headache
   B. Cluster headache
   C. Migraine
   D. Trigeminal neuralgia

2. Which of the following is a diagnostic feature of migraine with aura?
   A. Aura lasting <5 minutes
   B. Aura occurring after headache
   C. Gradual onset of reversible focal neurological symptoms
   D. Persistent neurological deficit

3. The drug of choice for acute moderate to severe migraine is:
   A. Propranolol
   B. Amitriptyline
   C. Sumatriptan
   D. Topiramate

4. Which drug is commonly used for migraine prophylaxis?
   A. Ibuprofen
   B. Propranolol
   C. Tramadol
   D. Ergotamine

Active space

5. A 38-year-old woman with history of migraine now has daily dull headaches for 6 months. She consumes analgesics almost daily. Most likely diagnosis?
A. Chronic migraine
B. Tension-type headache
C. Medication-overuse headache
D. Cluster headache

6. Medication-overuse headache is most commonly associated with overuse of:
A. Antihypertensives
B. NSAIDs, triptans, opioids
C. Antidepressants
D. Antiepileptics

7. The most important step in management of medication-overuse headache is:
A. Increase analgesic dose
B. Start opioids
C. Gradual withdrawal of overused drugs
D. Immediate imaging

8. A 60-year-old man complains of sudden, brief, electric shock–like facial pain triggered by chewing and brushing teeth.

Most likely diagnosis?

A. Migraine
B. Cluster headache
C. Trigeminal neuralgia
D. Temporal arteritis

Active space

9. The drug of choice for trigeminal neuralgia is:
A. Gabapentin
B. Carbamazepine
C. Amitriptyline
D. Sumatriptan

10. Which feature favors trigeminal neuralgia over migraine?
A. Unilateral pain
B. Associated nausea
C. Very short-lasting paroxysms
D. Photophobia

11. A 72-year-old woman presents with new-onset headache, scalp tenderness, jaw claudication, and visual blurring.Most likely diagnosis?
A. Migraine
B. Cluster headache
C. Temporal arteritis
D. Trigeminal neuralgia

Active space

CHAPTER 7: MOTOR NEURON DISEASES

What are motor neuron diseases?

Classification of motor neuron diseases:

Active space

Amyotrophic Lateral Sclerosis:

- Also known as________________________________

Risk factors:

- Age
- Genetics

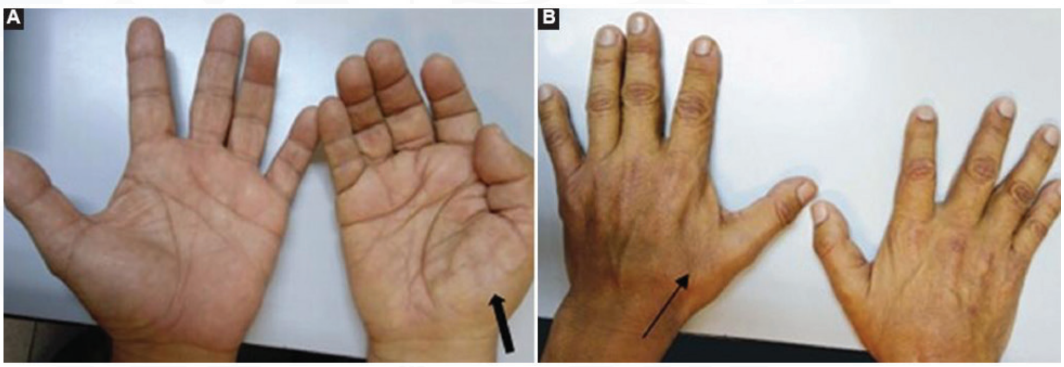
Active space

Clinical features:

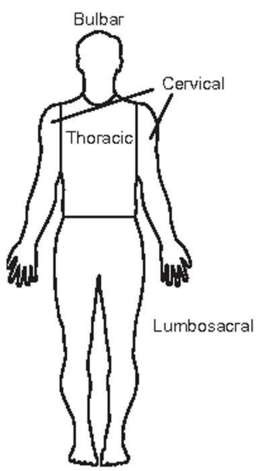
- Bulbar / Pseudobulabr palsy

&

- Progressive weakness of limbs



Diagnosis:



	Clinical presentation
Definite ALS	UMN and LMN signs in three regions (pictured left)
Probable ALS	UMN and LMN signs in two regions *AND* some UMN signs must be rostral to (above) LMN signs
Possible ALS	UMN and LMN signs in one region *OR* UMN signs alone in two or more regions *OR* UMN signs alone in two regions
Suspected ALS	LMN signs alone in two or more regions
All	Signs of degeneration spread within or between regions *AND* Lack of electrophysiological or neuroimaging evidence for other disorders which would explain observed degeneration

UMN signs: spasticity, hyperreflexivity *LMN signs: weakness, atrophy, fasciculation*

Active space

Management:

- Supportive care

- Pharmacotherapy

Active space

Workbook Questions

1. A 55-year-old man presents with progressive limb weakness, muscle wasting, brisk reflexes, and fasciculations. Sensory examination is normal. Which of the following best explains this constellation?
A. Primary muscle disease
B. Peripheral neuropathy
C. Combined upper and lower motor neuron degeneration
D. Disorder of neuromuscular junction

2. Which of the following is NOT a feature of amyotrophic lateral sclerosis?
A. Fasciculations
B. Hyperreflexia
C. Sensory loss
D. Progressive muscle weakness

3. A patient with suspected ALS shows tongue atrophy with fasciculations and spastic quadriparesis. This represents involvement of:
A. Only lower motor neurons
B. Only upper motor neurons
C. Both bulbar LMN and corticospinal UMN
D. Peripheral nerves

4. Which motor neuron disease is characterized by pure upper motor neuron involvement?
A. Progressive muscular atrophy
B. Amyotrophic lateral sclerosis
C. Primary lateral sclerosis
D. Spinal muscular atrophy

Active space

5. A 48-year-old man has progressive asymmetric distal limb weakness, muscle wasting, and absent reflexes without spasticity or bulbar involvement. The most likely diagnosis is:
A. ALS
B. Primary lateral sclerosis
C. Progressive muscular atrophy
D. Myasthenia gravis

6. Which of the following drugs has been shown to modestly improve survival in ALS?
A. Levodopa
B. Riluzole
C. Pyridostigmine
D. Interferon-β

Active space

CHAPTER 8: MENINGITIS

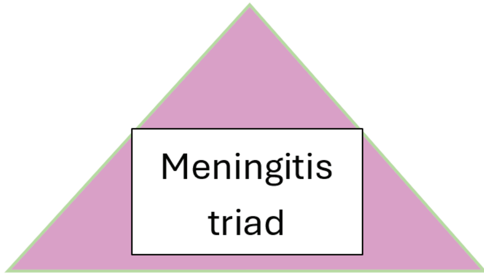
Definition:

Classification of meningitis :

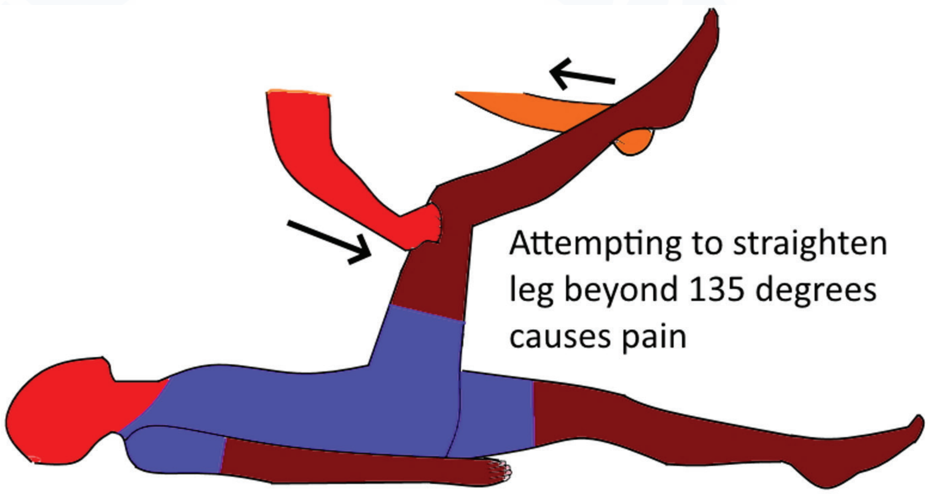
1. Bacterial
   - M/C/C ....
   - 2nd M/C/C ....
   - Other causes
2. Viral
3. Fungal
4. Tubercular meningitis

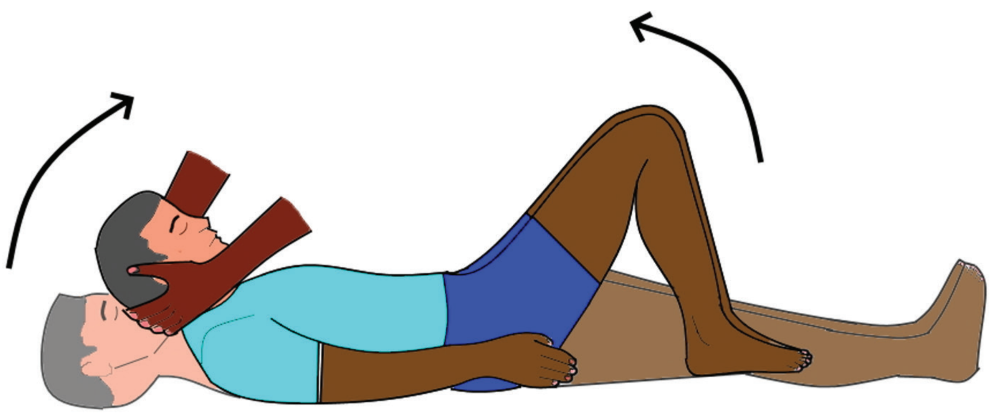
Active space

Clinical features:



Examination findings::





Active space

Approach to a case of meningitis:

1. Lumbar puncture for CSF

Note: Prerequisite ....

Parameter	Bacterial	Viral	Fungal	Tubercular
Cell count				
Differential count				
Protein				
Glucose				
Additional findings				

2. Empirical antibiotics:

a. First dose should be given .... hour of presentation

b. Can we delay antibiotics if LP is delayed?

Active space

Empirical choice of antibiotics	
Scenario	Antibiotics
Adults aged < 55 years with suspected / confirmed community acquired meningitis	
Adults aged > 55 years with suspected / confirmed community acquired meningitis	
Hospital acquired / health care associated meningitis	
Meningitis in patients with penetrating head injuries	

Active space

3. Role of steroids?

Active space

Workbook Questions

1. A 22-year-old student presents with acute onset fever, headache, vomiting, neck stiffness, and altered sensorium. CSF shows high opening pressure, neutrophilic pleocytosis, low glucose, and high protein. The most likely diagnosis is:
A. Viral meningitis
B. Tubercular meningitis
C. Acute bacterial meningitis
D. Fungal meningitis

2. Which organism is the most common cause of community-acquired bacterial meningitis in adults?
A. Neisseria meningitidis
B. Haemophilus influenzae
C. Streptococcus pneumoniae
D. Listeria monocytogenes

3. A patient with suspected meningitis has papilledema and focal neurological deficits. The next best step is:
A. Immediate lumbar puncture
B. Start steroids only
C. CT brain before lumbar puncture
D. Delay all treatment

4. A 35-year-old man presents with subacute headache, fever, cranial nerve palsy, and altered sensorium. CSF shows lymphocytic pleocytosis, very high protein, and low glucose. Most likely diagnosis?
A. Viral meningitis
B. Pyogenic meningitis
C. Tubercular meningitis
D. Carcinomatous meningitis

Active space

5. Adjunctive dexamethasone therapy in bacterial meningitis is most beneficial in infection caused by:
A. Neisseria meningitidis
B. Streptococcus pneumoniae
C. Listeria monocytogenes
D. Escherichia coli

6. Which organism is classically associated with meningitis in neonates, elderly, and immunocompromised patients?
A. Streptococcus pneumoniae
B. Neisseria meningitidis
C. Listeria monocytogenes
D. Haemophilus influenzae

7. Which of the following CSF findings favors fungal meningitis?
A. Neutrophils, normal glucose
B. Lymphocytes, moderately low glucose
C. Very low protein, normal glucose
D. Predominant eosinophils

Active space

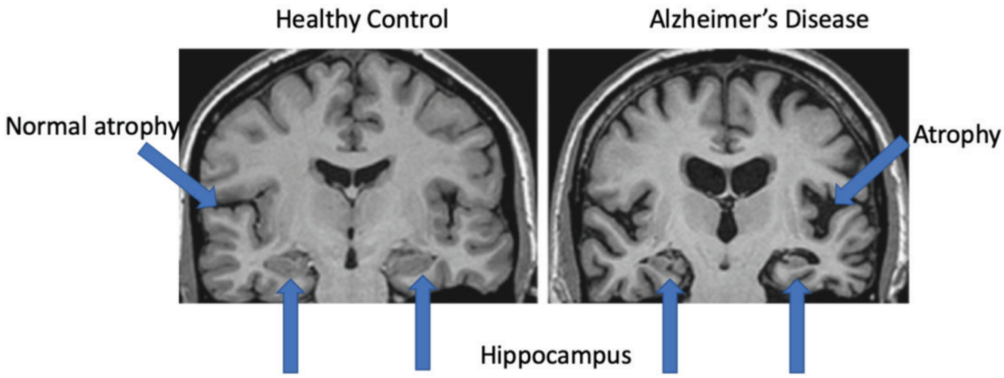
CHAPTER 9: MOVEMENT DISORDERS

Active space

CHAPTER 10: NEURODEGENERATIVE DISORDERS

Alzheimers Disease:

Active space



Diagnosis of AD:

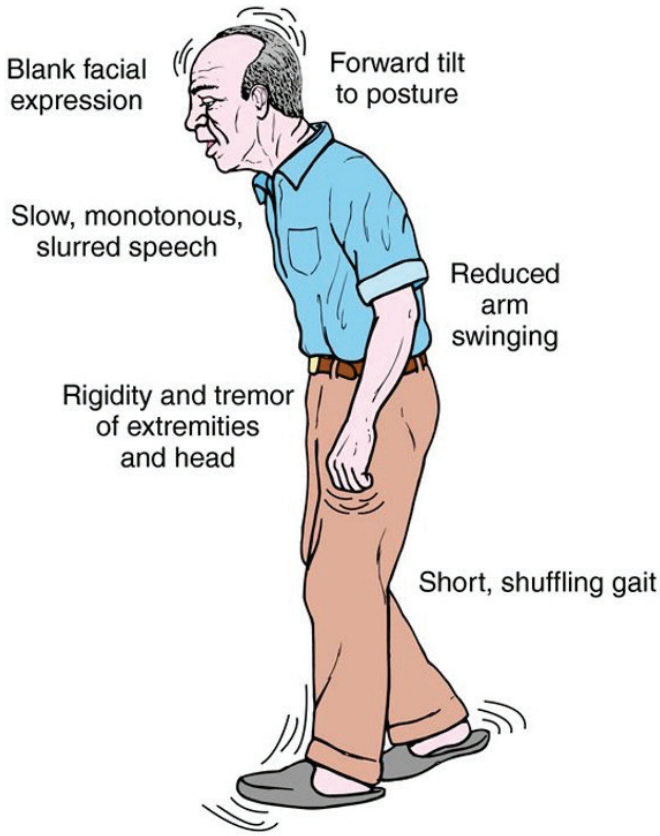
Active space

Management of AD:

Active space

Parkinson's Disease:

Active space



Diagnosis of PD:

Active space

MIST

Management of PD:

Active space

Workbook Questions

1. A 62-year-old man presents with resting tremor, rigidity, bradykinesia, and postural instability. Which of the following is the core pathological abnormality in Parkinson's disease?
A. Loss of GABAergic neurons in striatum
B. Loss of dopaminergic neurons in substantia nigra pars compacta
C. Demyelination of corticospinal tracts
D. Degeneration of Purkinje cells
Answer: B
☞ Dopamine depletion in nigrostriatal pathway

2. Which of the following is NOT a typical feature of idiopathic Parkinson's disease?
A. Resting tremor
B. Bradykinesia
C. Early autonomic dysfunction
D. Cogwheel rigidity
Answer: C
☞ Early autonomic failure suggests atypical parkinsonism

3. A patient with Parkinson's disease has tremor predominantly at rest that decreases with voluntary movement. This tremor is best described as:
A. Intention tremor
B. Postural tremor
C. Resting tremor
D. Action tremor
Answer: C

4. Which clinical feature helps differentiate Parkinson's disease from essential tremor?
A. Bilateral involvement at onset
B. Head tremor
C. Improvement with alcohol
D. Presence of bradykinesia
Answer: D
☞ Bradykinesia is essential for Parkinson's diagnosis

Active space

5. A 68-year-old patient with Parkinson's disease develops early falls, vertical gaze palsy, and poor response to levodopa. Most likely diagnosis?

A. Idiopathic Parkinson's disease
B. Multiple system atrophy
C. Progressive supranuclear palsy
D. Corticobasal degeneration

Answer: C

☞ PSP → early falls + gaze palsy

6. Which of the following drugs provides symptomatic benefit and modest disease-modifying effect in Parkinson's disease?

A. Selegiline
B. Trihexyphenidyl
C. Amantadine
D. Bromocriptine

Answer: A

☞ MAO-B inhibitors have mild neuroprotective effect (exam perspective)

7. A Parkinson's patient on long-term levodopa therapy develops involuntary choreiform movements at peak dose. This complication is known as:

A. Wearing-off phenomenon
B. On-off phenomenon
C. Peak-dose dyskinesia
D. Akinesia paradoxica

Answer: C

Active space

8. Which movement disorder is characterized by sudden, brief, shock-like involuntary jerks?

A. Chorea
B. Dystonia
C. Myoclonus
D. Athetosis

Answer: C

9. A patient presents with asymmetric limb rigidity, apraxia, cortical sensory loss, and alien limb phenomenon. The most likely diagnosis is:

A. Parkinson's disease
B. Progressive supranuclear palsy
C. Corticobasal degeneration
D. Multiple system atrophy

Answer: C

☞ MAO-B inhibitors have mild neuroprotective effect (exam perspective)

10. Which of the following is a cardinal motor feature required for diagnosis of Parkinson's disease?

A. Tremor
B. Rigidity
C. Postural instability
D. Bradykinesia

Answer: D

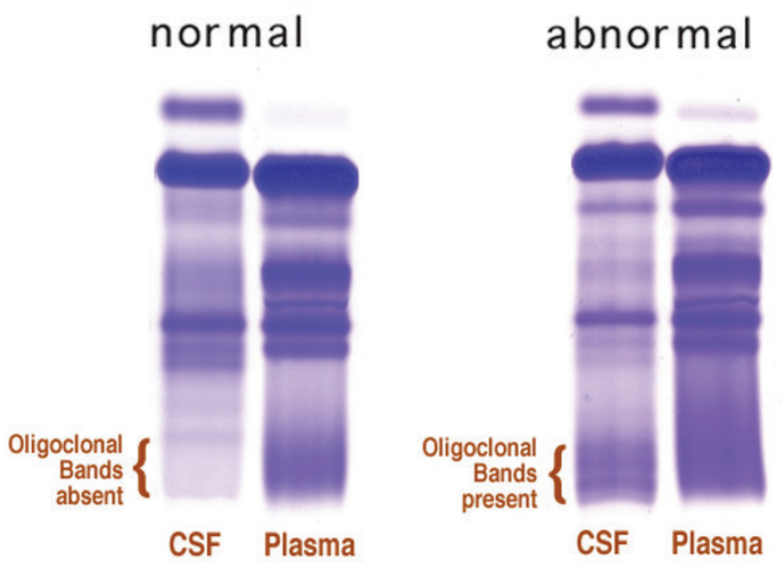
☞ Bradykinesia is mandatory for diagnosis

Active space

CHAPTER 11: MULTIPLE SCLEROSIS

Active space

Oligoclonal Bands in CSF



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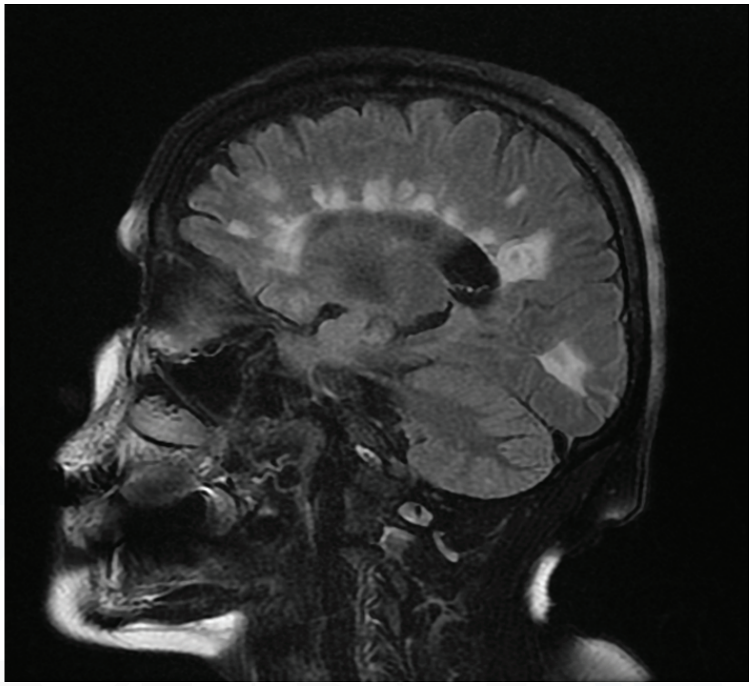


Active space

Diagnosis of MS:

Management of MS:

Active space

Active space

Active space

Workbook Questions

1. A 26-year-old woman presents with acute unilateral painful visual loss and impaired color vision. Fundoscopy is normal. The most likely diagnosis is:
A. Central retinal artery occlusion
B. Optic neuritis
C. Papilledema
D. Anterior ischemic optic neuropathy

Answer: B
☞ Painful visual loss with normal fundus → retrobulbar optic neuritis

2. Which of the following is the most characteristic pathological feature of multiple sclerosis?
A. Axonal degeneration only
B. Gray matter necrosis
C. Immune-mediated demyelination with relative axonal preservation
D. Neuronal inclusion bodies
Answer: C

3. A patient presents with recurrent episodes of neurological deficits separated by time and affecting different CNS locations. This clinical pattern is best described as:
A. Progressive course
B. Relapsing-remitting course
C. Secondary progressive course
D. Primary progressive course

Answer: B
☞ Most common MS subtype

Active space

4. Which MRI finding is most suggestive of multiple sclerosis?

A. Basal ganglia calcification

B. Periventricular white-matter lesions perpendicular to ventricles

C. Diffuse cerebral atrophy

D. Pontine hemorrhage

Answer: B

☞ Dawson's fingers

5. Which CSF finding supports the diagnosis of multiple sclerosis?

A. Neutrophilic pleocytosis

B. Low glucose

C. Oligoclonal IgG bands

D. High lactate

Answer: C

6. A patient with MS develops worsening neurological deficits over 6 months without distinct relapses after an initial relapsing–remitting phase. This course is termed:

A. Primary progressive MS

B. Progressive relapsing MS

C. Secondary progressive MS

D. Benign MS

Answer: C

7. Which of the following is a poor prognostic factor in multiple sclerosis?

A. Female sex

B. Optic neuritis at onset

C. Sensory symptoms at onset

D. Primary progressive disease

Answer: D

Active space

8. Which drug is commonly used for acute relapse management in multiple sclerosis?
A. Interferon-β
B. Glatiramer acetate
C. High-dose intravenous methylprednisolone
D. Azathioprine

Answer: C

9. Which symptom is characteristic of MS and worsens with heat exposure?
A. Lhermitte's sign
B. Uhthoff's phenomenon
C. Romberg sign
D. Babinski sign

Answer: B
☞ Transient worsening with heat

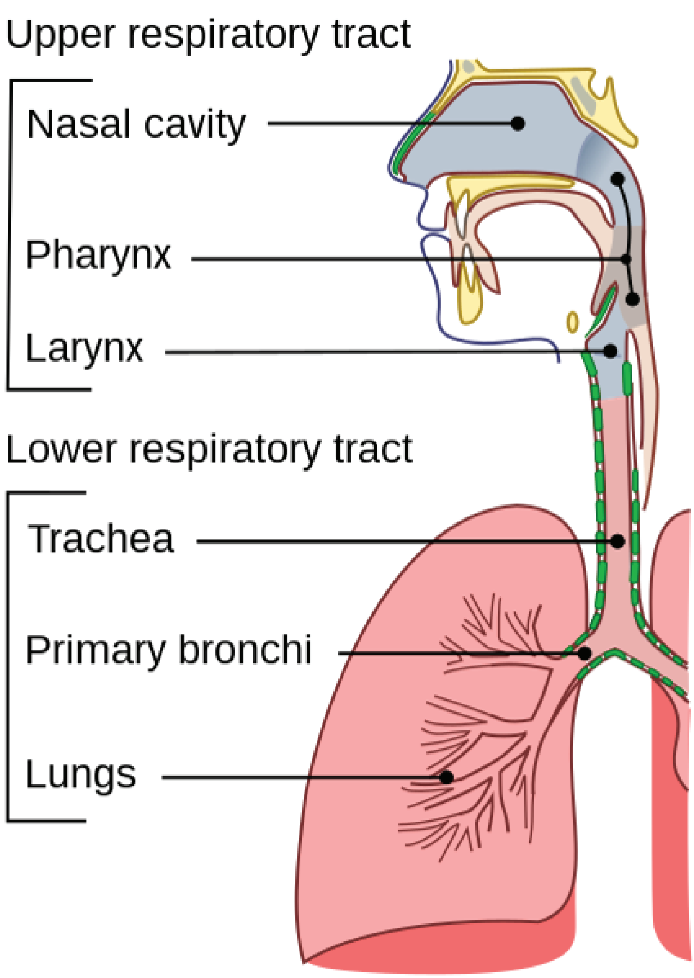
10. Which of the following neurological features is uncommon in multiple sclerosis and suggests an alternative diagnosis?
A. Optic neuritis
B. Internuclear ophthalmoplegia
C. Early bowel and bladder involvement
D. Sensory disturbances

Answer: C

Active space

CLINICAL PULMONOLOGY

CHAPTER 1: BASIC FUNCTIONAL ANATOMY OF THE RESPIRATORY SYSTEM



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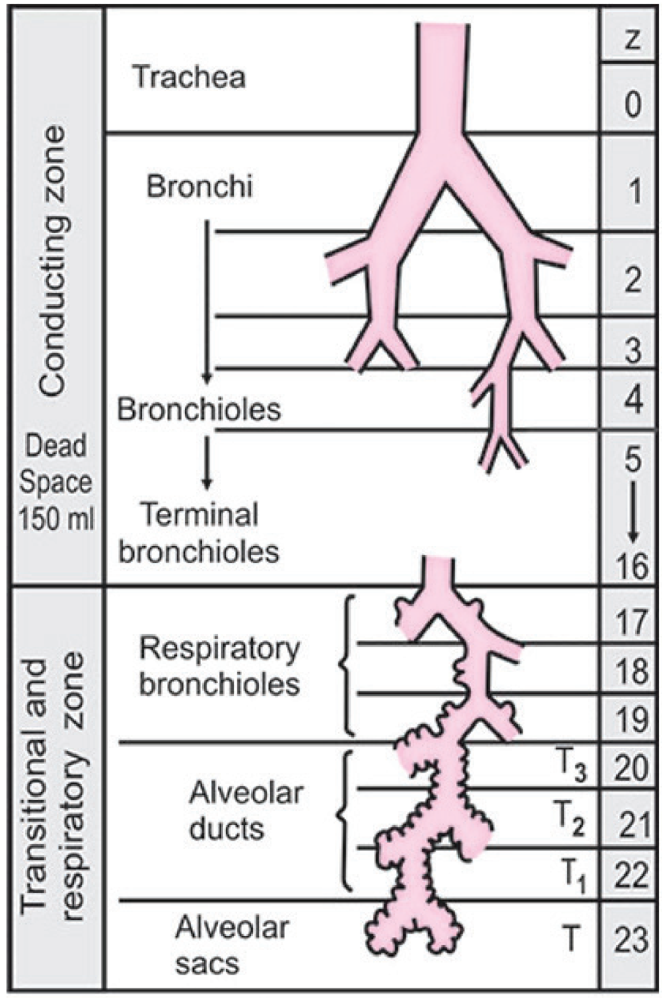


Divisions of
Tracheo-bronchial tree-23 divisions

Active space

CHAPTER 2: CLASSIFICATION OF RESPIRATORY DISORDERS BASED ON PFT

How to differentiate restrictive and obstructive disorders using PFT?

- FEV1=Volume of air exhaled after a deep maximal inspiration, forcefully in 1st second of exhalation
- FVC= Total volume of air exhaled after a deep maximal inspiration, forcefully

FEV1	FVC	FEV1/FVC ratio	Type of Respiratory Disorder
			Normal
			Obstructive Disorder
			Restrictive Disorder

Active space

How to differentiate between asthma and COPD?

DLCO :

- Tests integrity of ....
- Decreased in disorders affecting ....
- Examples-
- DLCO may be increased in ....

....

Active space

CHAPTER 3: BRONCHIAL ASTHMA

Definition:

Obstructive airway disorder characterized by .... airway obstruction.

Classification:

	Intrinsic	Extrinsic
Triggers		
Predictability		
Age of onset		
Samter's triad		
Prognosis		

Active space

	Type 2	Non-type 2
Th cell involved		
Cytokines involved		
Effector cells involved		

Biomarkers of Type 2 Asthma:

Clinical Features of Asthma:

- Episodic shortness of breath, wheezing and cough
- Auscultation-

Active space

Diagnosis:

- IOC-
- FEV1/FVC
- BDR test
- Alternative to PFT

Active space

Management of asthma:

Two types of inhaled medications-

- Reliver → used to relieve an asthma attack which has already set in
- Maintenance → used to prevent asthma attacks
- Preferred treatment →

Active space

Making the choice of first line inhaler as per GINA guidelines:

Active space

Management of asthma:
Patient refractory to MD-MART:

Treatment options -

- HD MART
- Adding LAMA
- Biologicals
  - Omalizumab
  - Dupilimumab
  - Mepolimumab
  - Reslizumab
  - Benralizumab
  - Tazepelumab

Active space

Active space

Active space

Active space

CHAPTER 4: COPD

Definition:

Causes for COPD:

- M/C/C
- Pack years= No of .... cigarettes smoked/day X number of years smoked
- M/C/C for COPD in non-smoking females-
- Genetics -.. deficiency

  (increased risk of emphysema)

Clinical features of COPD:

1. Chronic bronchitis

   Also called as ....

   Productive cough for .... for .... years.

   Symptoms : cough with sputum, cyanosis

2. Emphysema

Active space

Also called as ....

Irreversible dilatation and destruction of ....

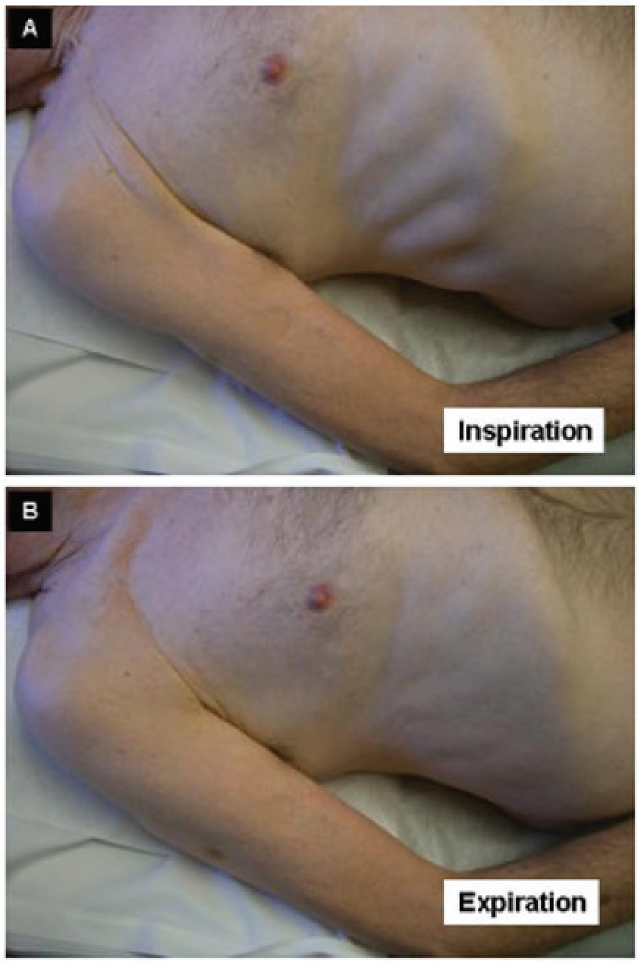
....

Clinical features :

- .... shaped chest
- Tripod position
- Pursed lip breathing
- Dahl sign – hyperpigmentation of ....

Active space

Active space



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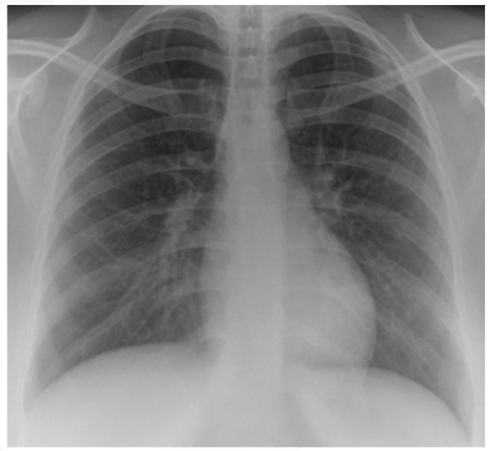
MIST

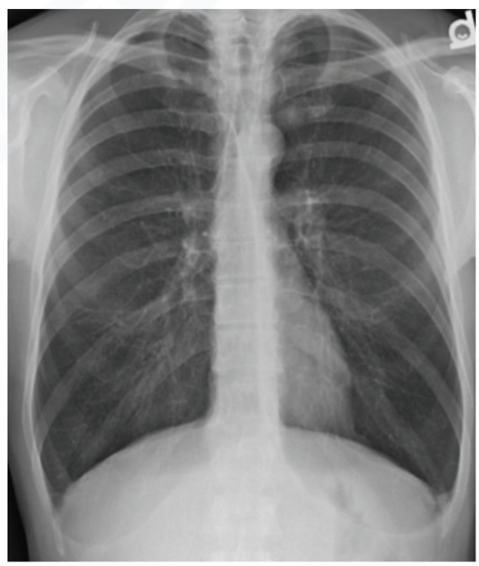
Diagnosis of COPD:

- PFT-

- DLCO-

- CXR-





Active space

GOLD staging of COPD:

- GOLD 1 - mild: FEV1 ≥80% predicted
- GOLD 2 - moderate: 50% ≤ FEV1 <80% predicted
- GOLD 3 - severe: 30% ≤ FEV1 <50% predicted
- GOLD 4 - very severe: FEV1 <30% predicted.

Management of COPD according to GOLD guidelines:

Vaccines recommended:

Active space

Supportive care:

Active space

Initial treatment:

Medications for refractory cases:

Ancillary treatments:

Active space

CHAPTER 5: INTERSTITIAL LUNG DISEASES (ILDS)

Definition: Chronic diffuse parenchymal lung disorders

Classification:

Idiopathic pulmonary fibrosis (IPF)

- M/C type of ILD
- Risk factors –
  - Gender
  - Age
  - Smoking
  - Genetics

✦ Clinical features-

- Shortness of breath
- Chronic cough
- Auscultation

✦ Diagnosis-

- IOC-
- UIP Pattern

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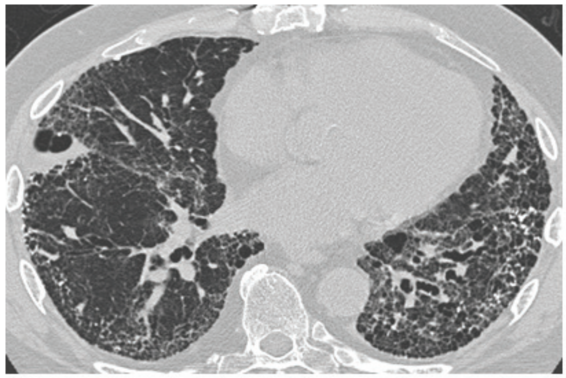


What do we get on PFT in ILD patients?

Management of IPF patients:

- 1st line
- 2nd line
- 3rd line

Idiopathic Non-specific Interstitial Pneumonia (NSIP):

- Age
- Gender
- Smoking?

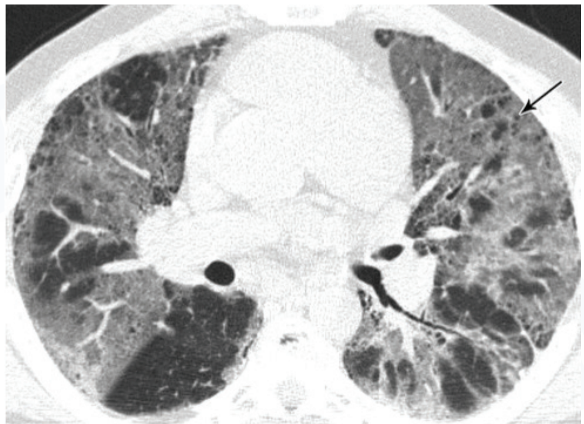
Clinical features – chronic dry cough & shortness of breath

- Auscultation –

Active space

Diagnosis:

- IOC-



Management:

- 1st line
- 2nd line

Active space

CHAPTER 6: RESPIRATORY FAILURE

What is respiratory failure?

Types of respiratory failure:

1. Type 1: Hypoxic respiratory failure

Examples:

- Pneumonia
- Pulmonary embolism
- ARDS
- Drowning

ABG:

- Pa02
- PaCO2

Active space

2. Type 2: Hypercapnic respiratory failure

Examples:

- COPD
- Opioids
- All type 1 causes can lead to type 2 respiratory failure

ABG:

- PaO2
- PaCo2

3. Type 3: Any of the above types of respiratory failure occurring in patients .... due to .... of the lungs.

4. Type 4: Any of types of respiratory failure occurring in patients ... .

Active space

Active space

CHAPTER 7: SARCOIDOSIS

Definition:

- Multiorgan inflammatory disease characterized by .... granulomas in the affected tissues.

Risk factors :

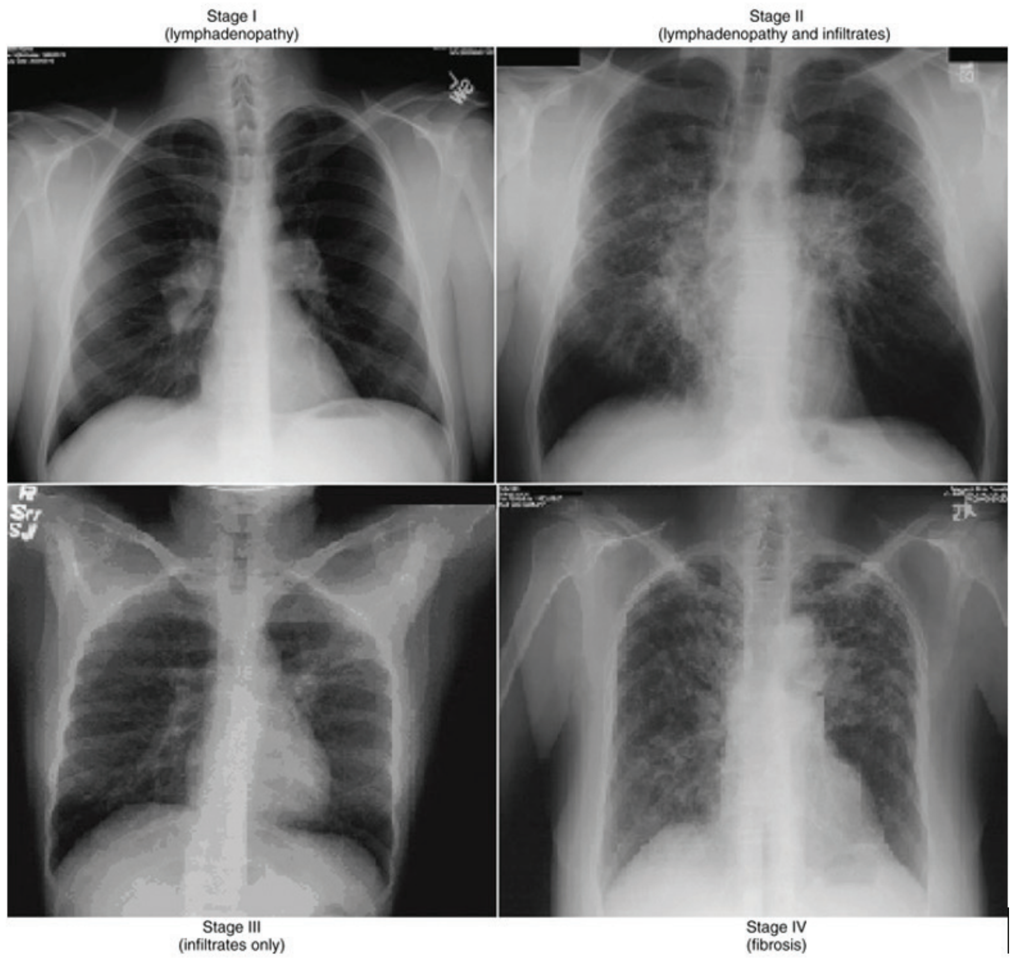
- Age –
- Gender –
- HLA association :
  - Increase risk - HLA DR 11,12,14,15,17.
  - Decrease risk - HLA DR 1, DR4, DQ *0202
- Ethnicity – African blacks, cardiac sarcoidosis, most common in Japanese.
- Occupation – agriculture / HCW / mining.

Clinical features :

- Lungs :
- Airway - Obstruction
- Parenchyma - Restrictive

Active space

Scadding staging of pulmonary sarcoidosis:



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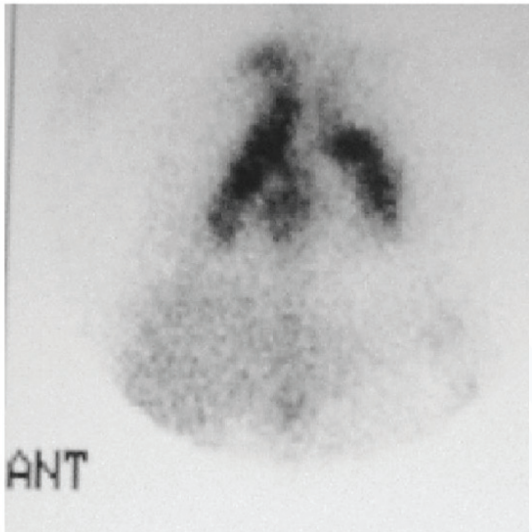
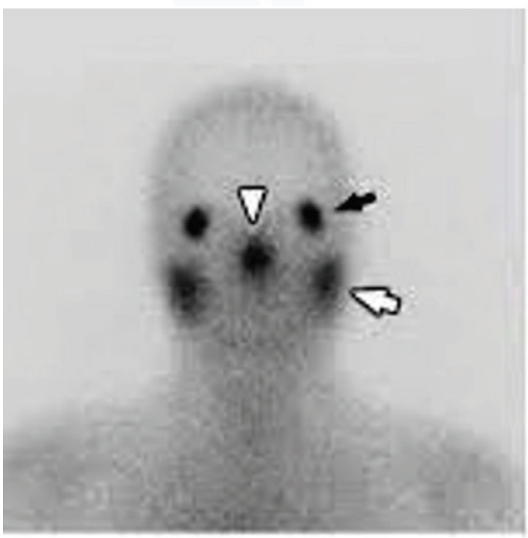
Investigations for pulmonary sarcoid:

- Nuclear scans:





Active space

- BAL fluid CD4/CD8 ratio ....

Cardiac manifestations :

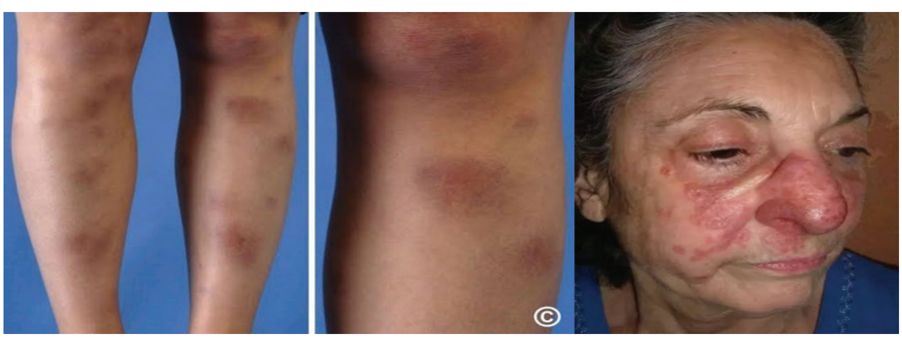
- Most common - conduction abnormalities (AV block).
- Myocarditis / cardiomyopathy → heart failure.
- Diagnosis – IOC is cardiac PET / Gd - MRI --Patchy increased tracer uptake in myocardium.

Neurosarcoidosis :

- Most common – ....
- Most common affected nerve – ....

Skin :

- 2nd Most commonly affected organ.
- 2 important manifestations- Lupus pernio & erythema nodosum.



Active space

Renal sarcoidosis :

- Most common:

Endocrine manifestations :

- Hypopituitarism
- Hypercalcemia & hypercalciuria,

Acute presentation of sarcoidosis :

1. Lofgren syndrome :
   - Fever
   - B/L arthritis (ankle)
   - B/L hilar lymphadenopathy
   - Erythema nodosum
   - Diagnosis: Visser's criteria :
     - Age : <40
     - Symptoms: duration <2 months B/L ankle arthritis
     - Erythema nodosum
   - Good Prognosis

Active space

2. Heerfordt syndrome / uveoparotid syndrome :

- Fever
- Parotitis : U/L or B/L.
- Anterior Uveitis
- Good prognosis

Diagnosis of sarcoidosis:

- Clinicoradiological picture consistent with sarcoidosis
- S. ACE levels > ....
- .... neg to take out TB.
- **Histopathological evidence** of non caseating granulomas / biopsy

  or

- classical MRI or PET evidence of cardiac sarcoidosis in biopsy from other sites not consistent.

Management :

- 50% resolves spontaneously.
- 2-3 months – observation
- For non resolving cases :
  - 1st line treatment : ....
  - 2nd line : ....

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CHAPTER 8: PLEURAL EFFUSION

What is pleural effusion?

Causes of pleural effusion :

1. Exudative :
   - Parapneumonic effusions
   - Tubercular effusions
   - Malignant effusions
   - CTD- related effusion(SLE/ RA)

2. Transudative :
   - CCF
   - CKD
   - CLD
   - Anemia
   - Decreased albumin

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Clinical features of pleural effusion :

Symptoms :

- Most common - ....
- Trepopnea ....
- Cough

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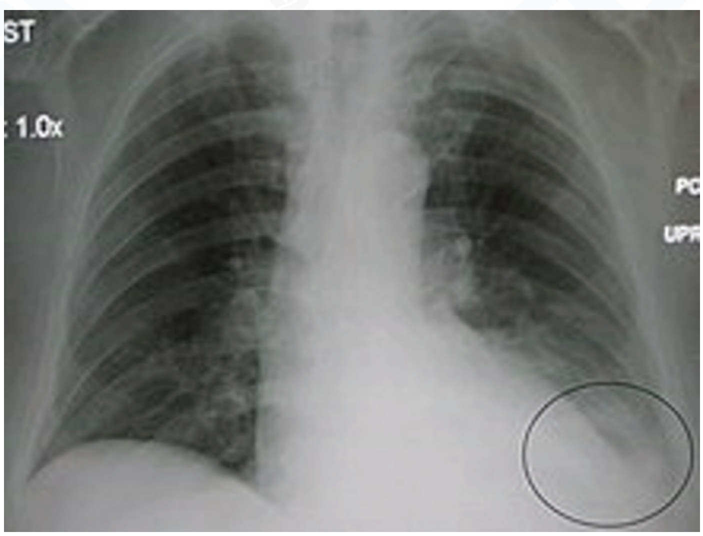
Examination findings :

- Inspection – If large trachea deviated to opposite rib spaces widening, movement of intercostal spaces decreased.
- Palpation – Confirm inspiratory finding
- Percussion – .... note
  - Elli's S-shaped curve
  - Skodaic resonance- a band of hyperresonance above the level of dullness.
- Auscultation :
  1. Breath sounds- ....
  2. Area above the effusions- .... breath sound

Diagnosis



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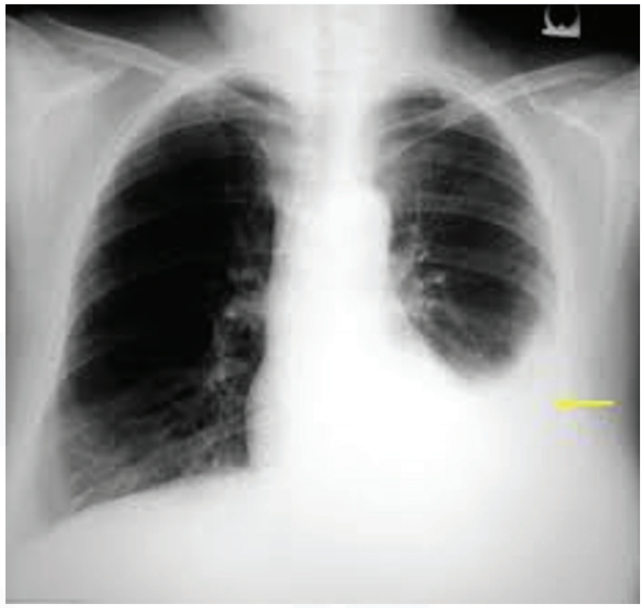

Approach to pleural effusion :

- IOC – Pleural fluid analysis

Light's criteria : To differentiate between exudative and transudative

Any 1/3 criteria +ve = Exudative

1. Pleural fluid protein/ serum protein ....
2. Pleural fluid LDH/ serum LDH ....
3. Pleural fluid LDH value .... of upper limit of normal.

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Pleural fluid glucose < 60 mg/dl :

- Parapneumonic effusion
- Rheumatoid arthritis
- Malignant effusion

How to identify hemothorax and chylothorax :

- Pleural fluid hematocrit .... of blood hematocrit = hemothorax
- Pleural fluid TGL... = chylothorax

How to identify tubercular effusion :

- Exudative
- ADA ....
- IOC- ....
- CBNAAT is not recommended because tubercular effusion is paucibacillary

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BONUS CHAPTER: ECG

Active space

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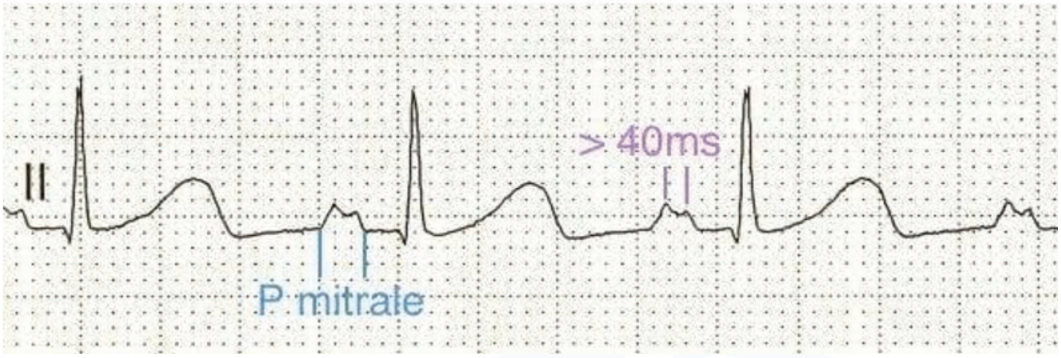

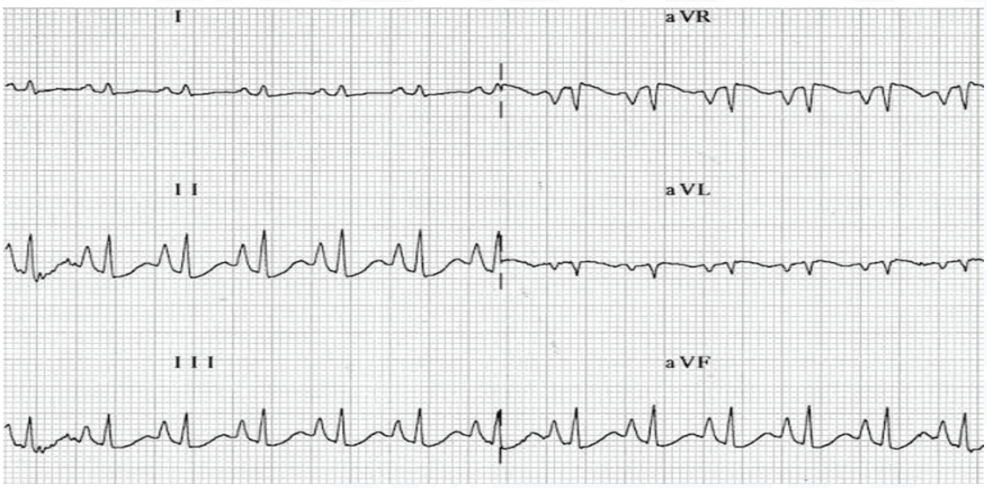

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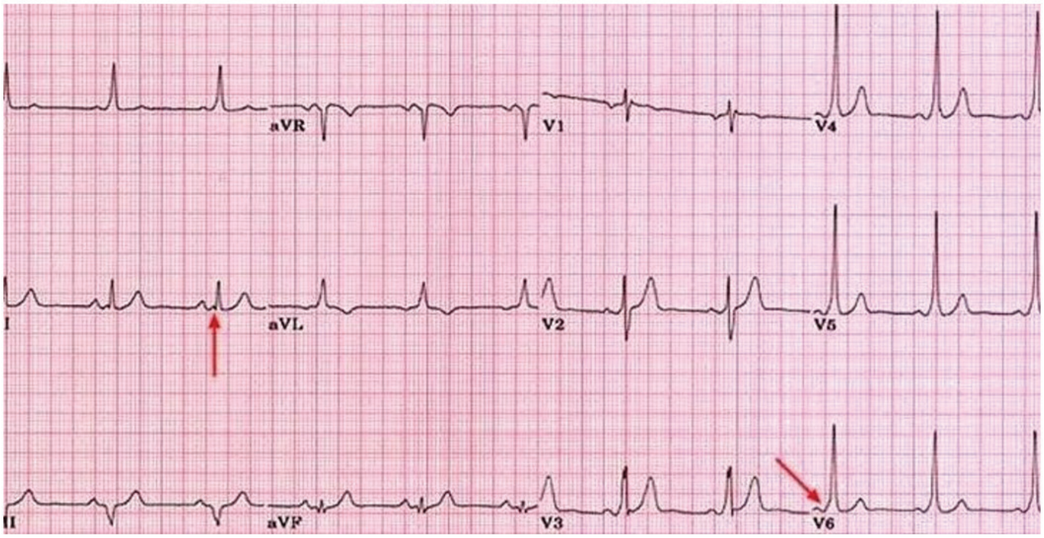

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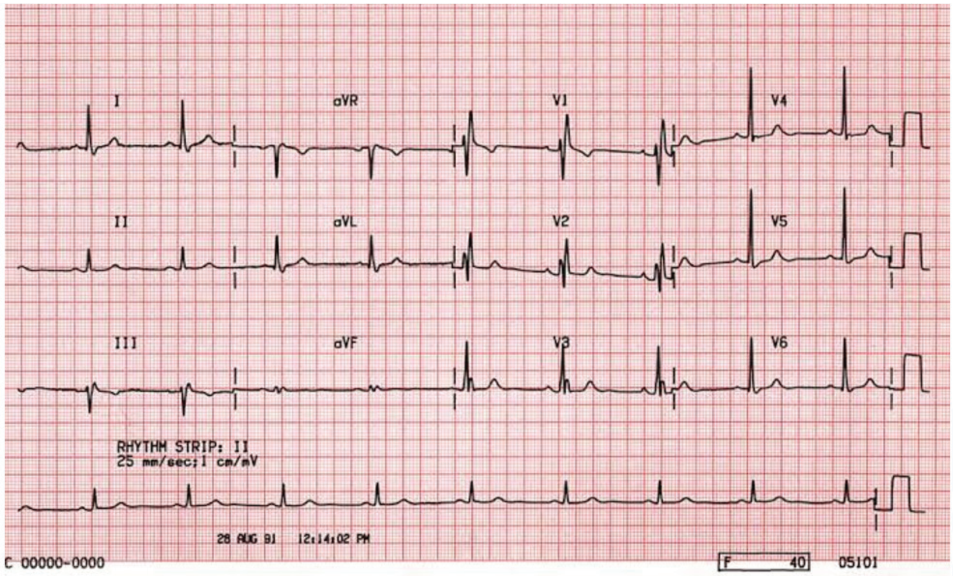

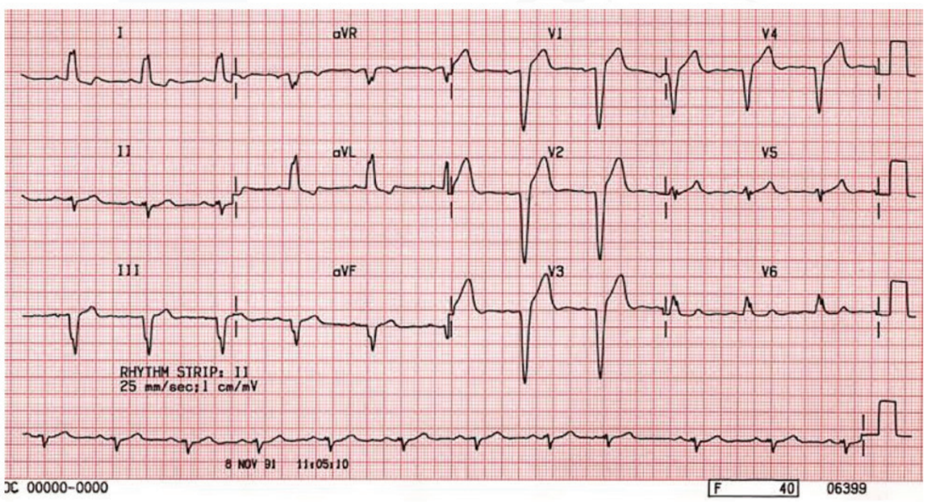

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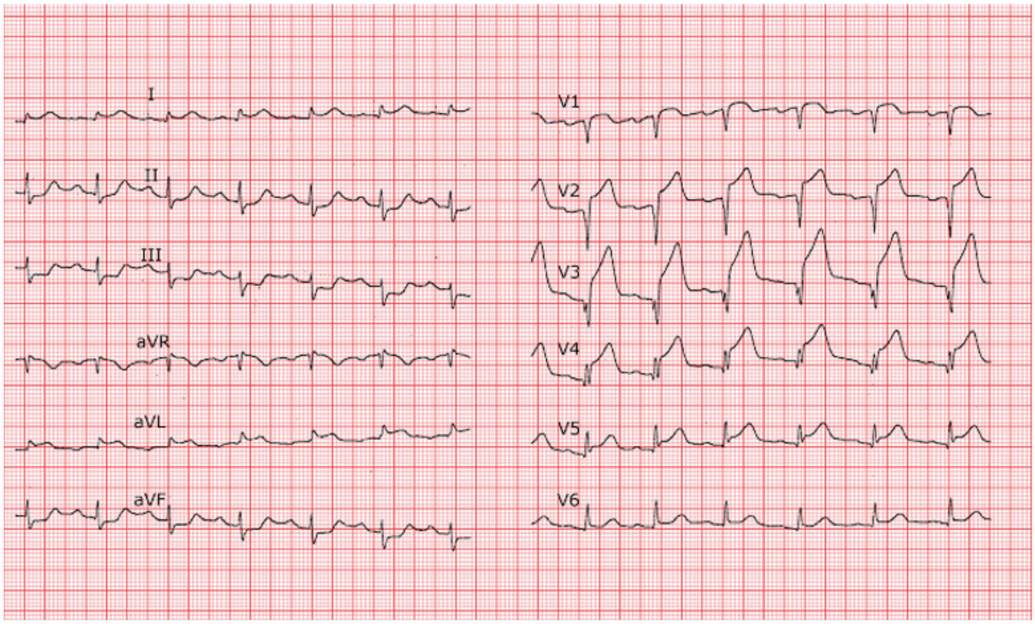

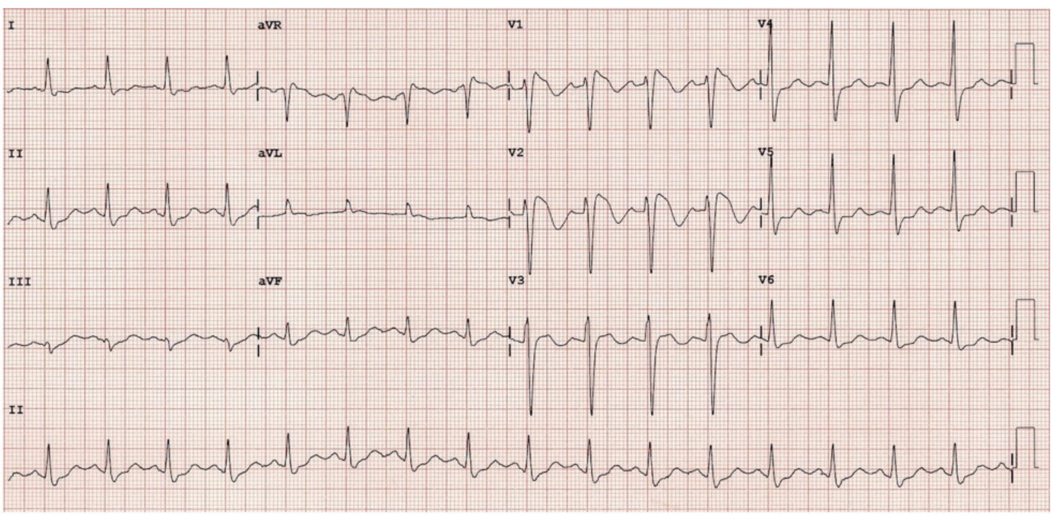

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First degree AV block

Second degree AV block (Mobitz I or Wenckebach)

Second degree AV block (Mobitz II)

Second degree AV block (2:1 block)

Third degree AV block with junctional escape

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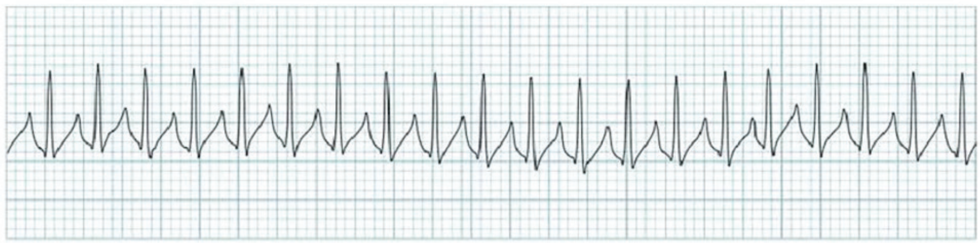

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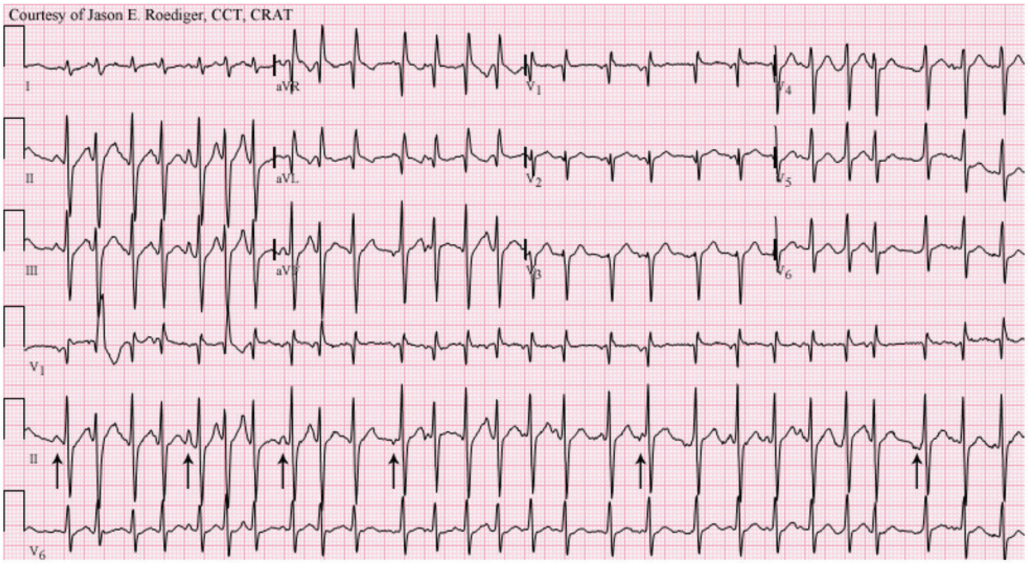

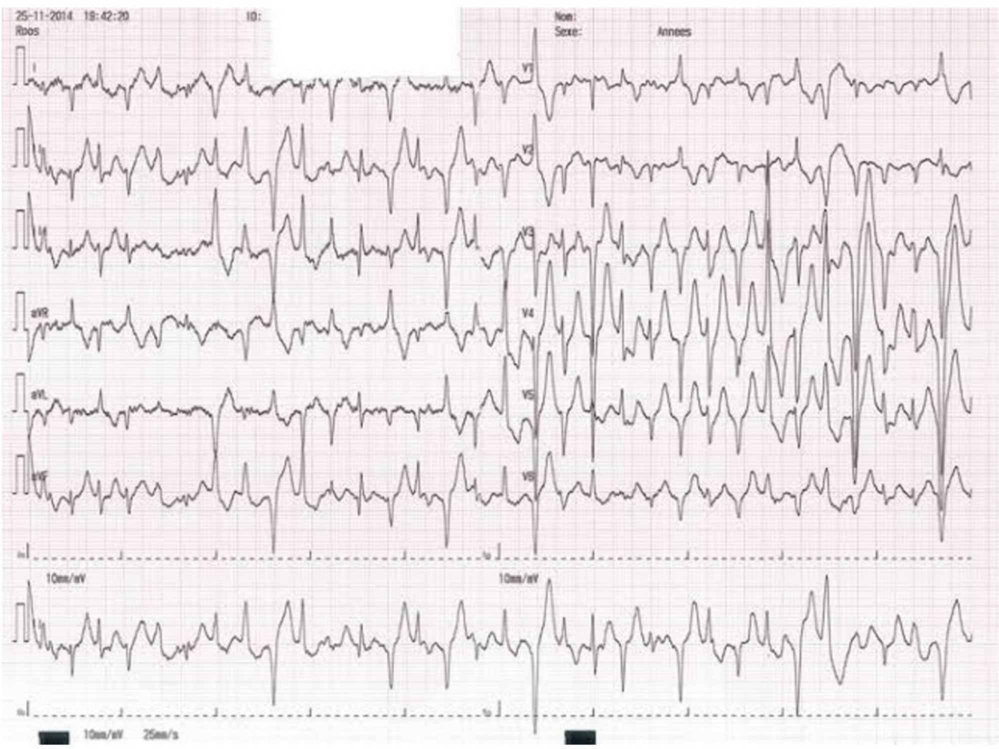

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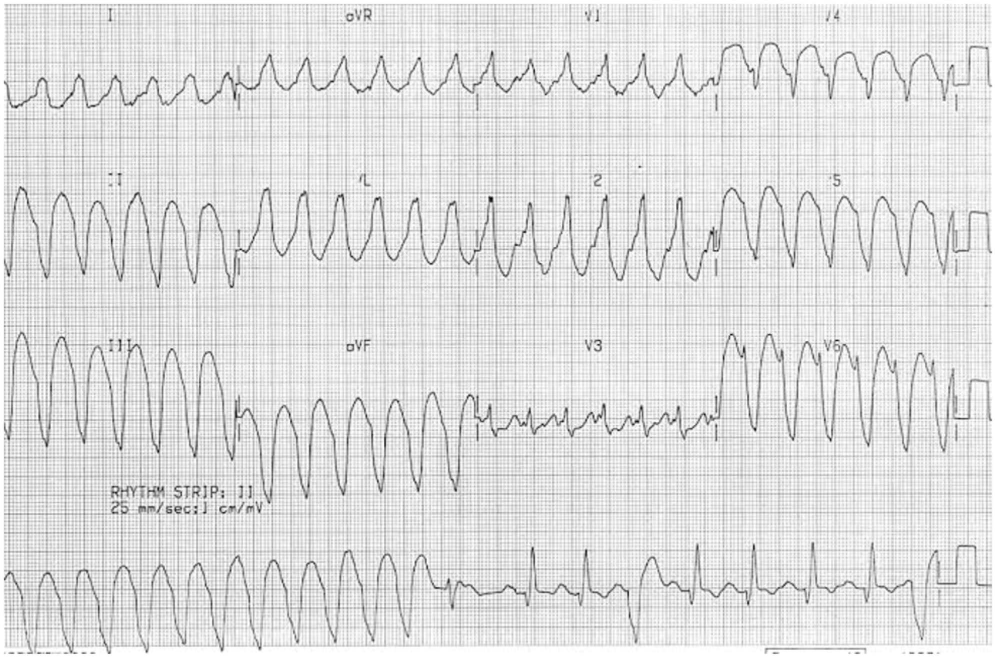

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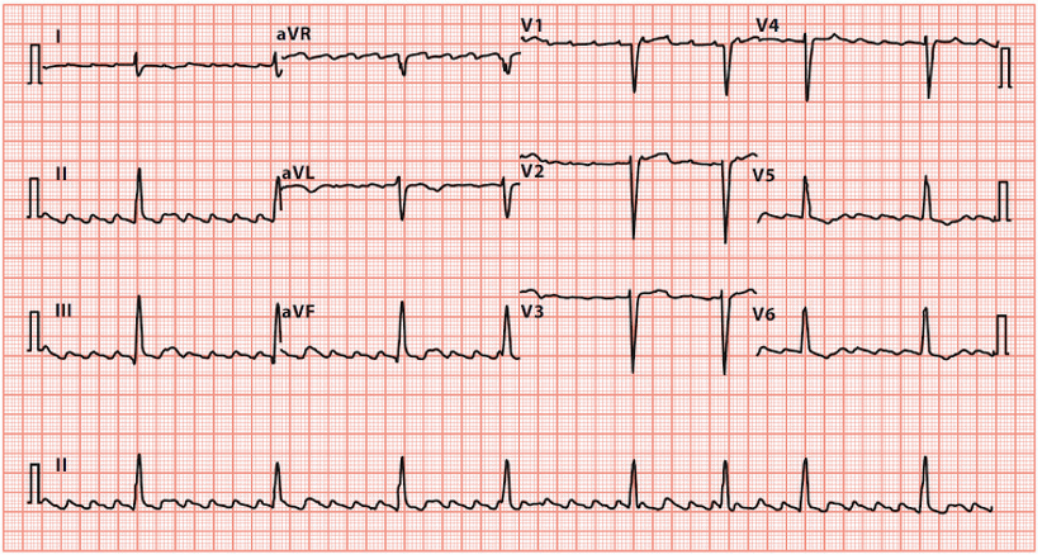

Active space

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